



**Systems and Software
Verification Laboratory**

MANCHESTER
1824

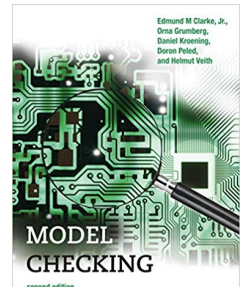
The University of Manchester

Detection of Software Vulnerabilities: Static Analysis

Lucas Cordeiro
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Static Analysis

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 - Office: 2.28
 - Office hours: 15-16 Tuesday, 14-15 Wednesday
- Textbook:
 - *Model checking* (Chapter 14)
 - *Software model checking*. ACM Comput. Surv., 2009
 - *The Cyber Security Body of Knowledge*, 2019
 - *Software Engineering* (Chapters 8, 13)



Motivating Example

- Functionality demanded increased significantly
 - Peer reviewing and testing

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- Multi-core processors with scalable shared memory / message passing
 - Static and dynamic verification

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```
void *threadA(void *arg) {  
    lock(&mutex);  
    x++;  
    if (x == 1) lock(&lock);  
    unlock(&mutex);  
    lock(&mutex);  
    x--;  
    if (x == 0) unlock(&lock);  
    unlock(&mutex);  
}
```

```
void *threadB(void *arg) {  
    lock(&mutex);  
    y++;  
    if (y == 1) lock(&lock);  
    unlock(&mutex);  
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void *threadB(void *arg) {  
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    if (y == 1) lock(&lock); (CS2)  
    unlock(&mutex);  
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Deadlock

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}
```

Intended learning outcomes

- Introduce **software verification** and **validation**

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 - The software should **conform to its specification**
- **Validation:** *"Are we building the right product"*
 - The software should do what the **user requires**
- Verification and validation must be applied at **each stage in the software process**
 - The **discovery of defects** in a system
 - The assessment of whether or not the system is **usable in an operational situation**

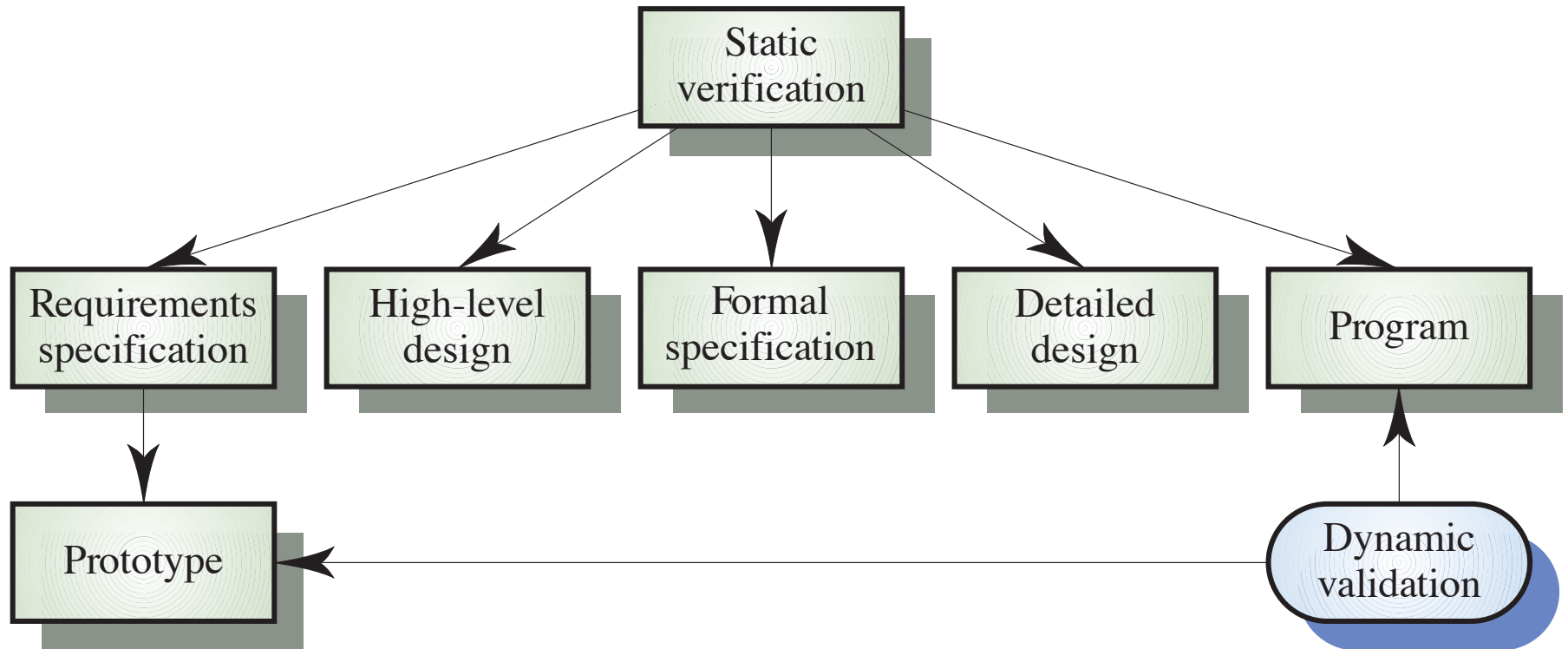
Static and Dynamic Verification

- **Software inspections** are concerned with the analysis of the static system representation to discover problems (**static verification**)
 - Supplement by **tool-based document and code analysis**
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 - **Code analysis** can **prove the absence of errors** but might be subject to **incorrect results**
- **Software testing** is concerned with exercising and observing product behaviour (**dynamic verification**)
 - The system is executed with **test data**
 - **Operational behaviour is observed**
 - Can reveal the presence of errors **NOT their absence**

Static and Dynamic Verification



Ian Sommerville. Software Engineering
(6th, 7th or 8th Edn) Addison Wesley

V & V planning

- **Careful planning** is required to get the most out of **dynamic and static verification**
 - Planning should start **early in the development process**
 - The plan should identify the **balance between static and dynamic verification**

V & V planning

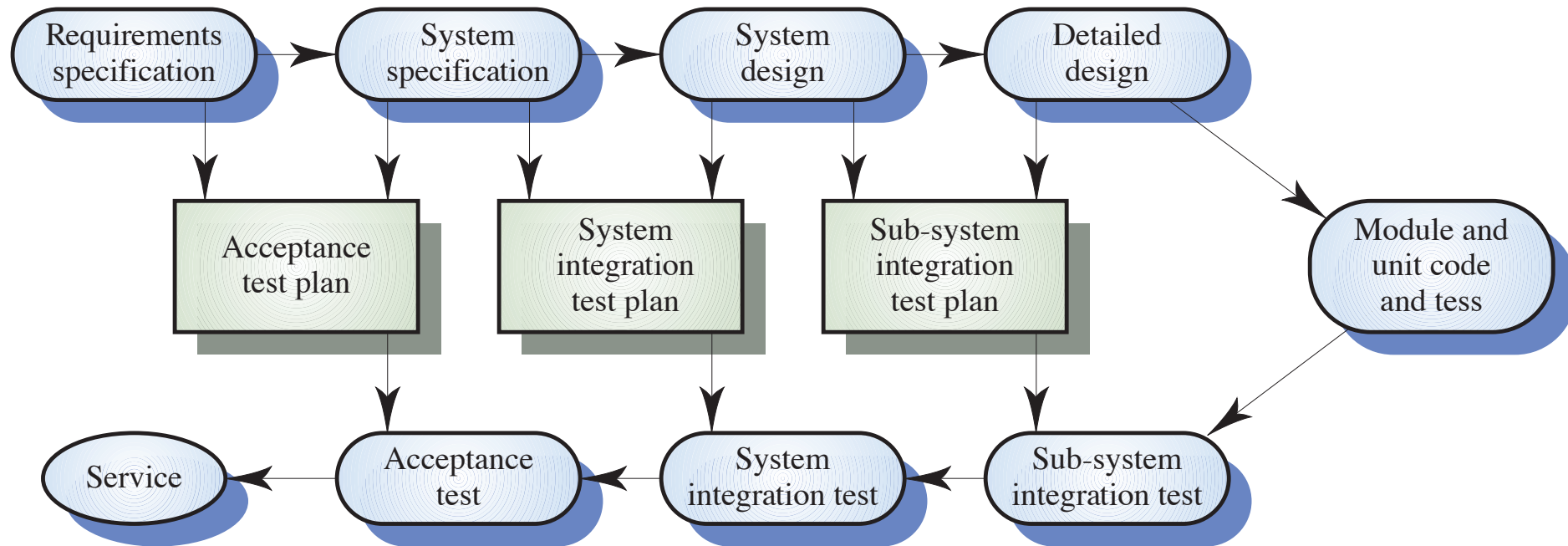
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V & V planning depends **on system's purpose, user expectations and marketing environment**

The V-model of development



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 - A detection technique is **complete** for a given category, if any vulnerability it finds is an actual vulnerability
 - An incomplete detection technique may have **false positives**, i.e., it may detect issues that do not turn out to be actual vulnerabilities

Detection of Vulnerabilities

- Achieving **soundness** requires reasoning about **all executions** of a program (usually an infinite number)
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 - This can be done by static checking of the program code while making suitable abstractions of the executions
- Achieving **completeness** can be done by performing actual, **concrete executions** of a program that are witnesses to any vulnerability reported
 - The analysis technique has to come up with concrete inputs for the program that triggers a vulnerability
 - A typical dynamic approach is software testing: the tester writes test cases with concrete inputs and specific checks for the outputs

Detection of Vulnerabilities

Detection tools can use a **hybrid combination of static and dynamic analysis** techniques to achieve a good trade-off between **soundness and completeness**

Detection of Vulnerabilities

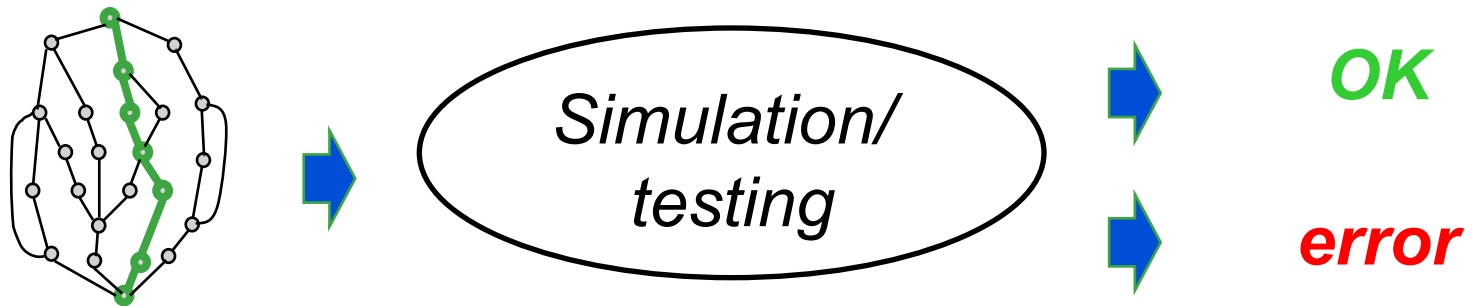
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Dynamic verification should be used in conjunction with **static verification** to provide **full code coverage**

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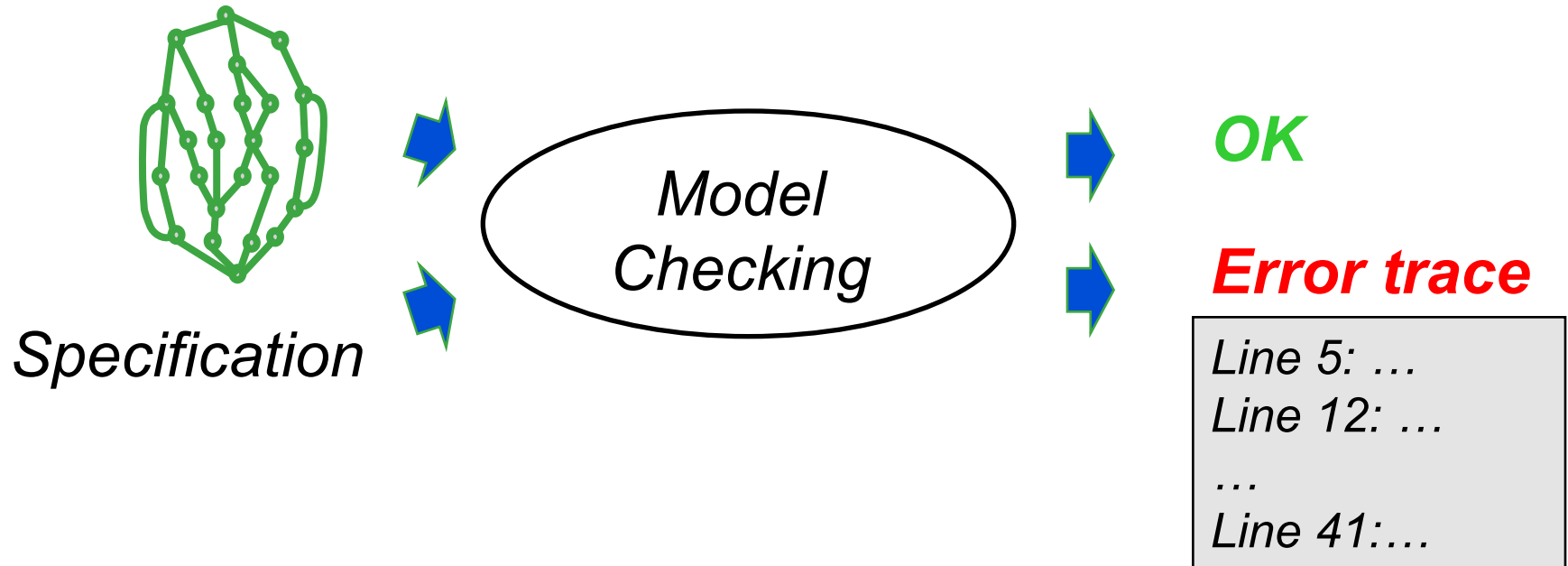
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Static analysis vs Testing/ Simulation



- **Checks only some of the system executions**
 - May miss errors
- A **successful execution** is an execution that **discovers one or more errors**

Static analysis vs Testing/ Simulation

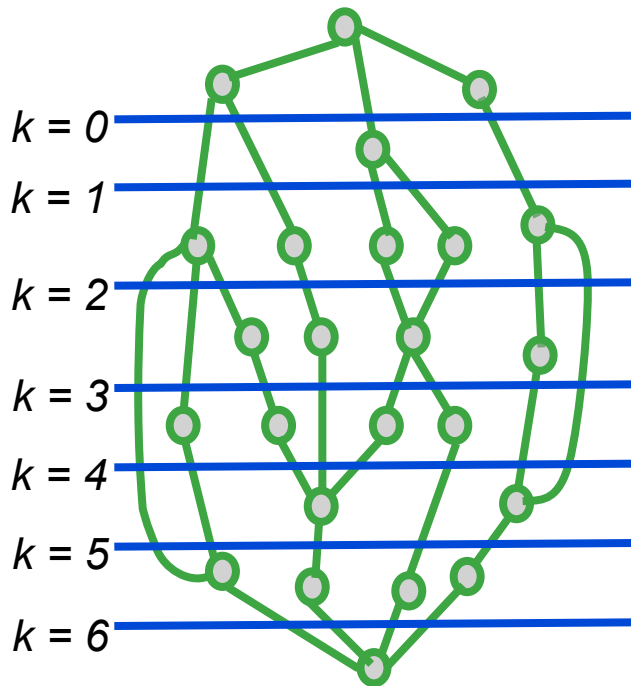


- **Exhaustively explores all executions**
- Report errors as **traces**
- May produce **incorrect results**

Avoiding state space explosion

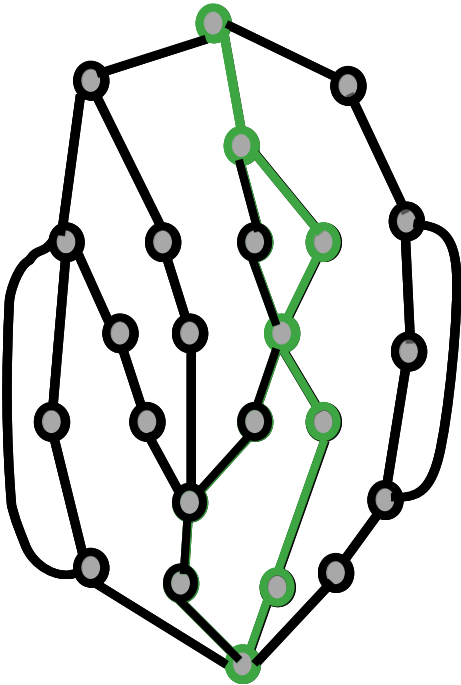
- Bounded Model Checking (BMC)
 - Breadth-first search (BFS) approach
- Symbolic Execution
 - Depth-first search (DFS) approach

Bounded Model Checking



- Bounded model checkers explore the state space in depth
- Can only prove correctness if all states are reachable within the bound

Symbolic Execution



- Symbolic execution explores all paths individually
- Can only prove correctness if all paths are explored

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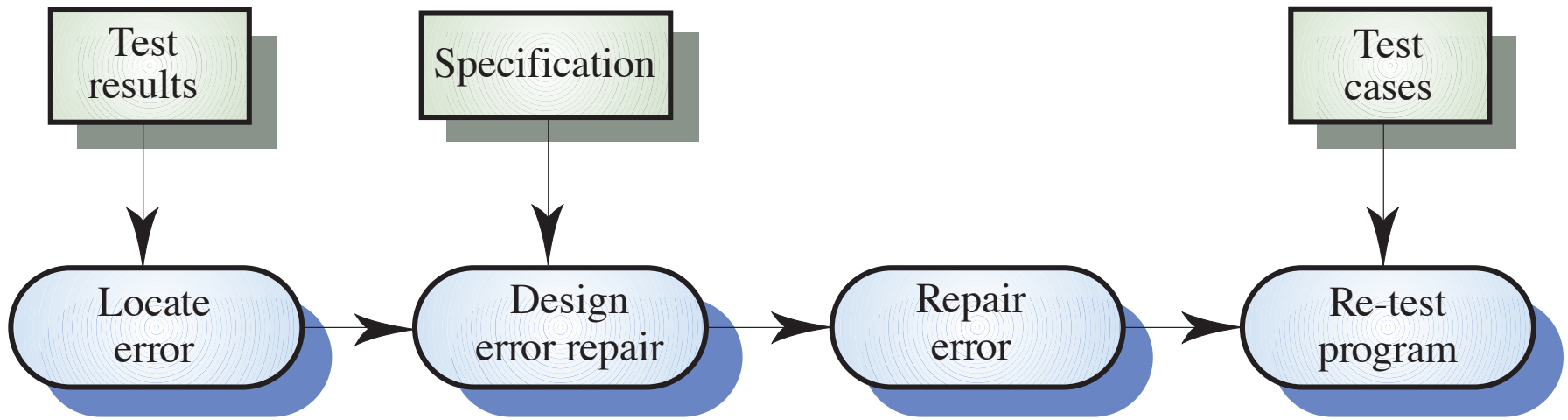
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- **Debugging** is concerned with two main tasks
 - **Locating and**
 - **Repairing these errors**
- Debugging involves
 - Formulating a hypothesis about program behaviour
 - Test these hypotheses to find the system error

The debugging process



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Intended learning outcomes

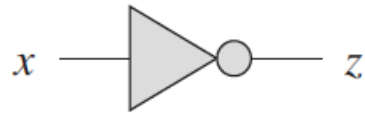
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Circuit Satisfiability

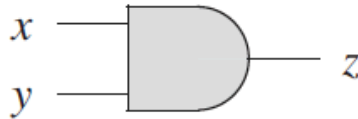
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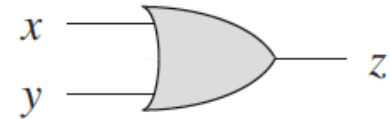
- A Boolean formula contains
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 - **Connectives**: $\wedge$ (**AND**), $\vee$ (**OR**), and $\neg$ (**NOT**)



x	$\neg x$
0	1
1	0



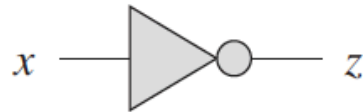
x	y	$x \wedge y$
0	0	0
0	1	0
1	0	0
1	1	1



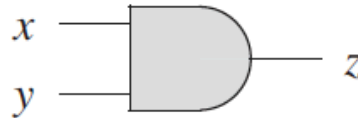
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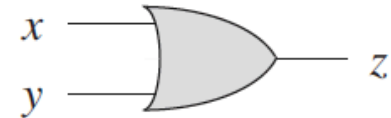
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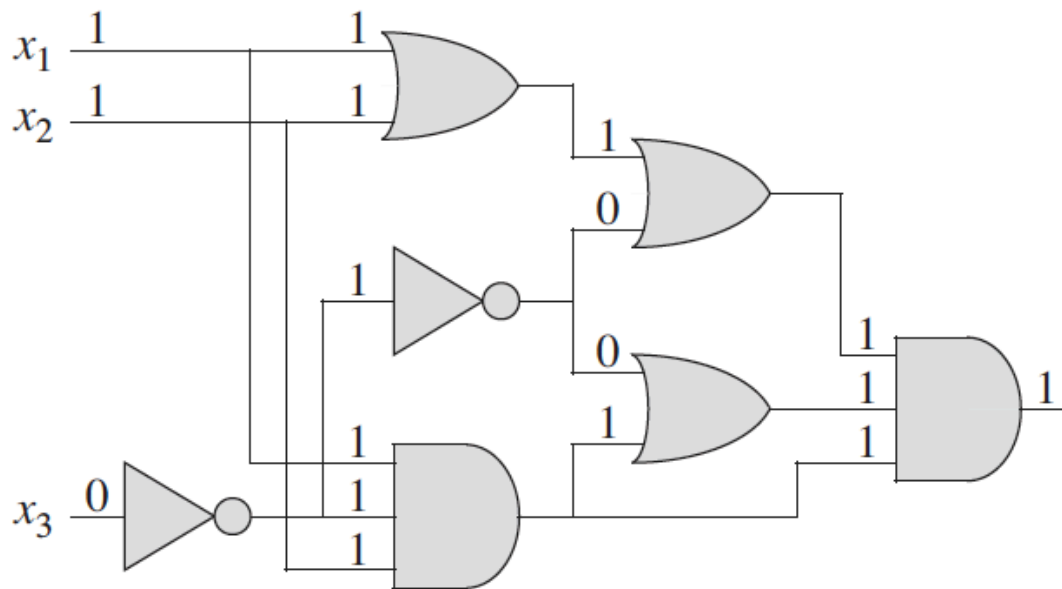


x	y	$x \vee y$
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0	1	1
1	0	1
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- A Boolean formula is **SAT** if there exists some assignment to its variables that **evaluates it to 1**

Circuit Satisfiability

- A **Boolean combinational circuit** consists of one or more **Boolean combinational elements** interconnected by **wires**



SAT: $\langle x_1 = 1, x_2 = 1, x_3 = 0 \rangle$

Circuit-Satisfiability Problem

- Given a **Boolean combinational circuit** of AND, OR, and NOT gates, is it **satisfiable**?

CIRCUIT-SAT = {<C> : C is a satisfiable Boolean combinational circuit}

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- When the **size of C is polynomial in k** , checking each one takes **$\Omega(2^k)$**
 - o Super-polynomial in the size of k

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- The SAT problem asks whether a given Boolean formula is satisfiable

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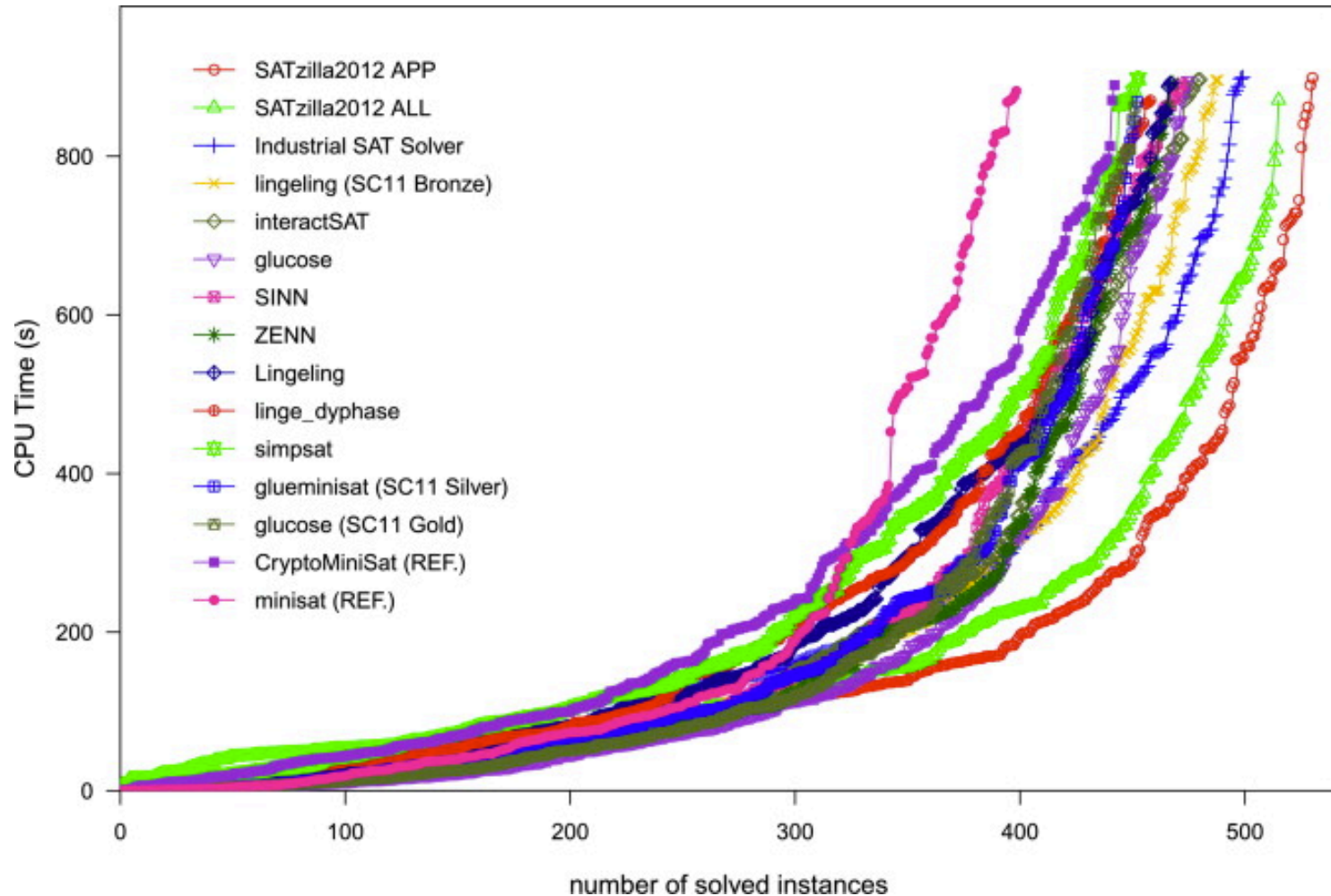
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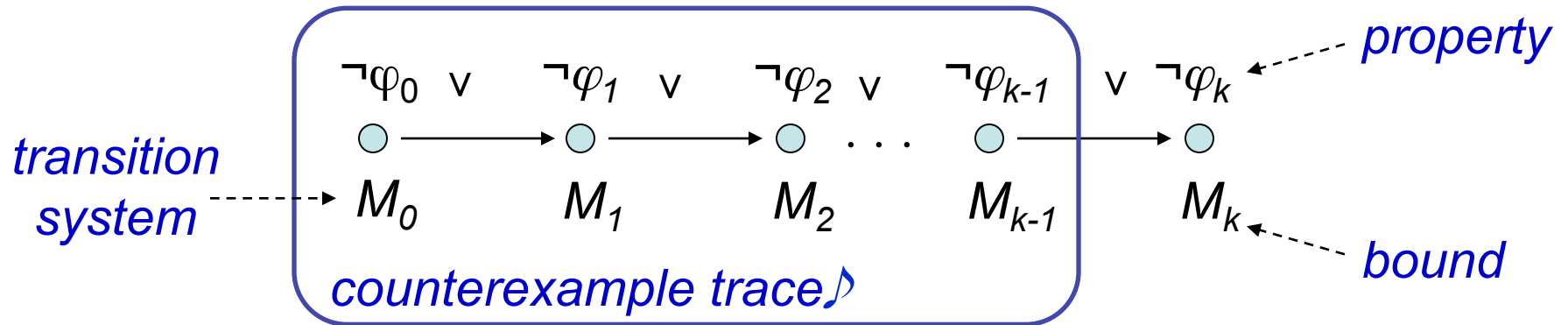
- $\Phi = 1$

SAT Competition



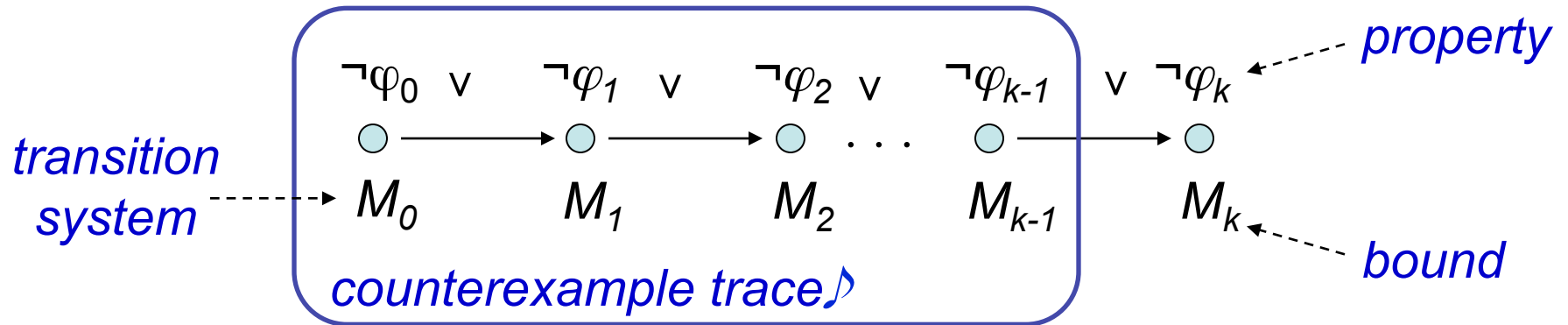
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Basic Idea: check negation of given property up to given depth



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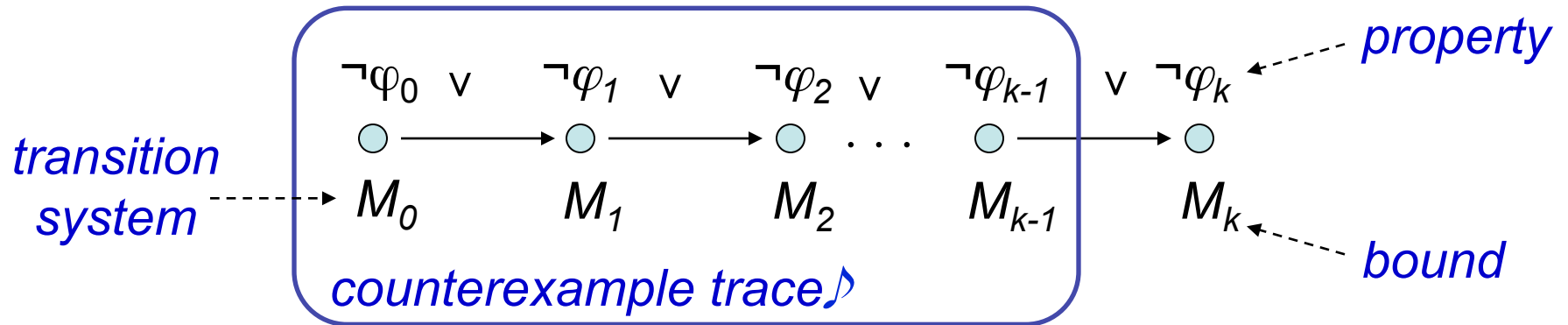
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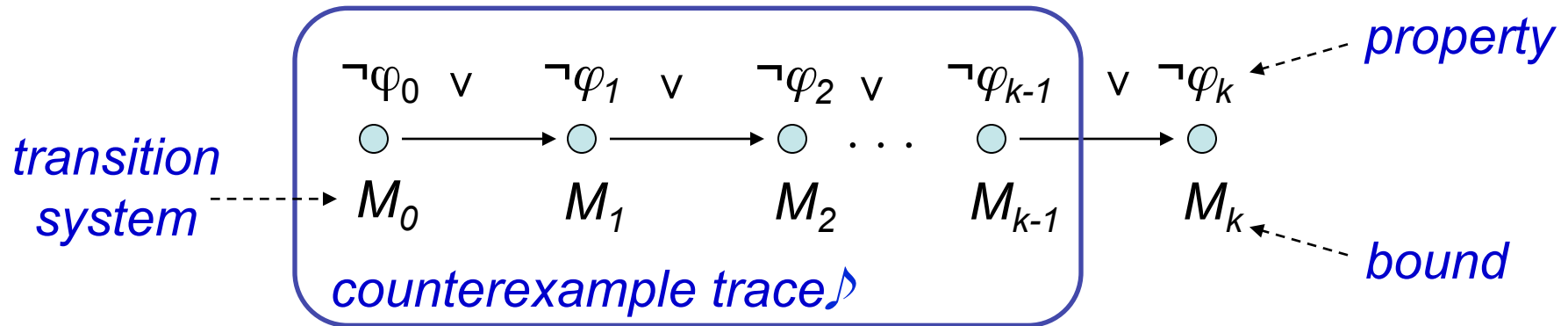
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- translated into verification condition ψ such that
 - ψ **satisfiable iff φ has counterexample of max. depth k**
- has been applied successfully to verify HW/SW systems

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Bit-vectors	$(b \gg i) \& 1 = 1$
Linear arithmetic	$(4y_1 + 3y_2 \geq 4) \vee (y_2 - 3y_3 \leq 3)$
Arrays	$(j = k \wedge a[k] = 2) \Rightarrow a[j] = 2$

Satisfiability Modulo Theories (1)

SMT decides the **satisfiability** of first-order logic formulae using the combination of different **background theories** (building-in operators)

Theory	Example
Equality	$x_1 = x_2 \wedge \neg (x_1 = x_3) \Rightarrow \neg (x_1 = x_3)$
Bit-vectors	$(b \gg i) \& 1 = 1$
Linear arithmetic	$(4y_1 + 3y_2 \geq 4) \vee (y_2 - 3y_3 \leq 3)$
Arrays	$(j = k \wedge a[k] = 2) \Rightarrow a[j] = 2$
Combined theories	$(j \leq k \wedge a[j] = 2) \Rightarrow a[i] < 3$

Satisfiability Modulo Theories (2)

- Given

- a decidable Σ -theory T
- a quantifier-free formula φ

φ is **T-satisfiable** iff $T \cup \{\varphi\}$ is satisfiable, i.e., there exists a structure that satisfies both formula and sentences of T

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φ is a **T-consequence of Γ** ($\Gamma \models_T \varphi$) iff every model of $T \cup \Gamma$ is also a model of φ

- Checking $\Gamma \models_T \varphi$ can be reduced in the usual way to checking the T-satisfiability of $\Gamma \cup \{\neg\varphi\}$

Satisfiability Modulo Theories (3)

- let **a** be an array, **b**, **c** and **d** be signed bit-vectors of width 16, 32 and 32 respectively, and let **g** be an unary function.

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$$\neq g(\text{SignExt}(b, 16) - c + 4) \wedge \text{SignExt}(b, 16) = c - 3 \wedge c + 1 = d - 4$$

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b' extends **b** to the signed equivalent bit-vector of size 32

$$\text{step1: } g(\text{select}(\text{store}(a, c, 12), b' + 3)) \neq g(b' - c + 4) \wedge b' = c - 3 \wedge c + 1 = d - 4$$

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replace **b'** by **c-3** in the inequality

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using facts about bit-vector arithmetic

$$\text{step 3: } g(\text{select}(\text{store}(a, c, 12), c)) \neq g(1) \wedge c - 3 = c - 3 \wedge c + 1 = d - 4$$

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The function g implies that for all x and y ,
if $x = y$, then $g(x) = g(y)$ (*congruence rule*).

step 5: SAT ($c = 5, d = 10$)

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if $x = y$, then $g(x) = g(y)$ (*congruence rule*).

step 5: $\text{SAT}(c = 5, d = 10)$

- SMT solvers also apply:
 - standard algebraic reduction rules
 - contextual simplification

$$\boxed{r \wedge \text{false} \mapsto \text{false}}$$

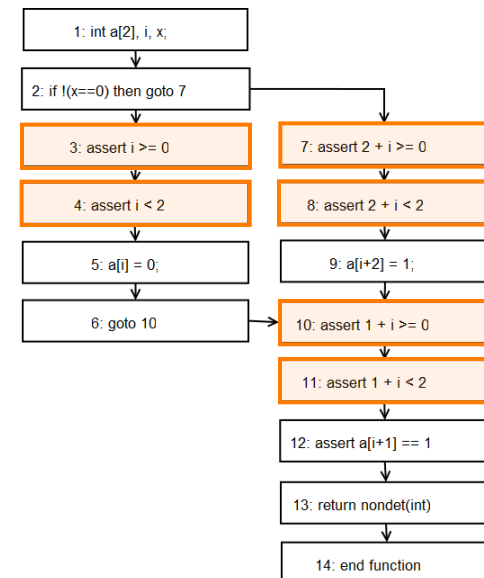
$$\boxed{a = 7 \wedge p(a) \mapsto a = 7 \wedge p(7)}$$

BMC of Software

- program modelled as state transition system
 - state: program counter and program variables
 - derived from control-flow graph
 - checked safety properties give extra nodes
- program unfolded up to given bounds
 - loop iterations
 - context switches
- unfolded program optimized to reduce blow-up
 - constant propagation
 - forward substitutions

} crucial

```
int main() {  
    int a[2], i, x;  
    if (x==0)  
        a[i]=0;  
    else  
        a[i+2]=1;  
    assert(a[i+1]==1);  
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```
 $g_1 = x_1 == 0$   
 $a_1 = a_0 \text{ WITH } [i_0 := 0]$   
 $a_2 = a_0$   
 $a_3 = a_2 \text{ WITH } [2+i_0 := 1]$   
 $a_4 = g_1 ? a_1 : a_3$   
 $t_1 = a_4[1+i_0] == 1$ 
```

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 } crucial
- front-end converts unrolled and optimized program into SSA
- extraction of constraints C and properties P
 - specific to selected SMT solver, uses theories
- satisfiability check of $C \wedge \neg P$

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$$C := \left[\begin{array}{l} g_1 := (x_1 = 0) \\ \wedge a_1 := \text{store}(a_0, i_0, 0) \\ \wedge a_2 := a_0 \\ \wedge a_3 := \text{store}(a_2, 2 + i_0, 1) \\ \wedge a_4 := \text{ite}(g_1, a_1, a_3) \end{array} \right]$$

$$P := \left[\begin{array}{l} i_0 \geq 0 \wedge i_0 < 2 \\ \wedge 2 + i_0 \geq 0 \wedge 2 + i_0 < 2 \\ \wedge 1 + i_0 \geq 0 \wedge 1 + i_0 < 2 \\ \wedge \text{select}(a_4, i_0 + 1) = 1 \end{array} \right]$$

Encoding of Numeric Types

- SMT solvers typically provide different encodings for numbers:
 - abstract domains (**Z**, **R**)
 - fixed-width bit vectors (`unsigned int`, ...)
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- majority of VCs solved faster if numeric types are modelled by abstract domains but possible loss of precision
- ESBMC supports both types of encoding and also combines them to improve scalability and precision

Encoding Numeric Types as Bitvectors

Bitvector encodings need to handle

- type casts and implicit conversions
 - arithmetic conversions implemented using word-level functions (part of the bitvector theory: Extract, SignExt, ...)
 - o different conversions for every pair of types
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 - $\text{unsigned_overflow} \Leftrightarrow (r - (r \bmod 2^w)) < 2^w$
 - define error literals to detect over- / underflow for other types
 - $\text{res_op} \Leftrightarrow \neg \text{overflow}(x, y) \wedge \neg \text{underflow}(x, y)$
 - o similar to conversions

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 - encode the integral (i) and fractional (f) parts
- **Binary encoding:** get a new bit-vector $b = i @ f$ with the same bitwidth before and after the radix point of a .

$$i = \begin{cases} \text{Extract}(b, n_b + m_a - 1, n_b) & : m_a \leq m_b \\ \text{SignExt}(\text{Extract}(b, t_b - 1, n_b), m_a - m_b) & : \text{otherwise} \end{cases} \quad \begin{array}{l} // m = \text{number of} \\ \text{bits of } i \end{array}$$

$$f = \begin{cases} \text{Extract}(b, n_b - 1, n_b - n_b) & : n_a \leq n_b \\ \text{Extract}(b, n_b, 0) @ \text{SignExt}(b, n_a - n_b) & : \text{otherwise} \end{cases} \quad \begin{array}{l} // n = \text{number of} \\ \text{bits of } f \end{array}$$

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- **Rational encoding:** convert a to a rational number

$$a = \begin{cases} \frac{\left(i * p + \left(\frac{f * p}{2^n} + 1 \right) \right)}{p} & : f \neq 0 \\ i & : \text{otherwise} \end{cases} \quad // p = \text{number of decimal places}$$

Floating-point SMT Encoding

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 - Floating-point arithmetic
 - Positive and negative infinities and zeroes
 - NaNs
 - Comparison operators
 - Five rounding modes: round nearest with ties choosing the even value, round nearest with ties choosing away from zero, round towards zero, round towards positive infinity and round towards negative infinity

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 - Z3: implements all operators
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 - o *fp.rem*: remainder: $x - y * n$, where n in $\mathbb{Z}$ is nearest to x/y
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 - *fp.rem*: remainder: $x - y * n$, where n in $\mathbb{Z}$ is nearest to x/y
 - *fp.fma*: fused multiplication and addition; $(x * y) + z$
- Both solvers offer non-standard functions:
 - *fp_as_ieeebv*: converts floating-point to bitvectors
 - *fp_from_ieeebv*: converts bitvectors to floating-point

How to encode Floating-point programs?

- Most operations performed at program-level to encode FP numbers have a **one-to-one conversion to SMT**
- Special cases being casts to boolean types and the fp.eq operator
 - Usually, cast operations are encoded using **extend/extract operation**
 - Extending floating-point numbers is non-trivial because of the format

```
int main()  
{  
    _Bool c;  
  
    double b = 0.0f;  
    b = c;  
    assert(b != 0.0f);  
  
    c = b;  
    assert(c != 0);  
}
```

Cast to/from booleans

- Simpler solutions:
 - Casting **booleans** to **floating-point numbers** can be done using an ite operator

```
(assert (= (ite |main::c|  
              (fp #b0 #b01111111111 #x0000000000000000)  
              (fp #b0 #b000000000000 #x0000000000000000))  
          |main::b|))
```

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If true, assign 1f to b

```
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Otherwise, assign 0f to b

Cast to/from booleans

- Simpler solutions:
 - Casting **floating-point numbers** to **booleans** can be done using an equality and one not:

```
(assert (= (not (fp.eq |main::b|  
                (fp #b0 #b000000000000 #x0000000000000000)))  
          |main::c|))
```

:note

"(fp.eq x y) evaluates to true if x evaluates to -zero and y to +zero, or vice versa. fp.eq and all the other comparison operators evaluate to false if one of their arguments is NaN."

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(assert (= (not (fp.eq |main::b|  
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```



true when the floating is not 0.0

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```

 *otherwise, the result is
false*

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Floating-point Encoding: Illustrative Example

```
int main()  
{  
    float x;  
    float y = x;  
    assert (x==y) ;  
    return 0;  
}
```

Floating-point Encoding: Illustrative Example

```
; declaration of x and y
(declare-fun |main::x| () (_ FloatingPoint 8 24))
(declare-fun |main::y| () (_ FloatingPoint 8 24))

; symbol created to represent a nondeterministic number
(declare-fun |nondet_symex::nondet0| () (_ FloatingPoint 8 24))

; Global guard, used for checking properties
(declare-fun |execution_statet::\guard_exec| () Bool)

; assign the nondeterministic symbol to x
(assert (= |nondet_symex::nondet0| |main::x|))

; assign x to y
(assert (= |main::x| |main::y|))

; assert x == y
(assert (let ((a!1 (not (=> true
                        (=> |execution_statet::\guard_exec|
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        (or a!1))))
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Variable declarations

```
; assert x == y
(assert (let ((a!1 (not (=> true
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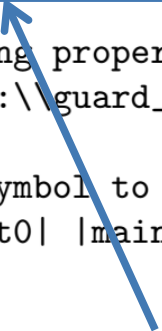
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```

**Nondeterministic symbol
declaration (optional)**



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Guard used to check
satisfiability

Floating-point Encoding: Illustrative Example

```
; declaration of x and y
(declare-fun |main::x| () (_ FloatingPoint 8 24))
(declare-fun |main::y| () (_ FloatingPoint 8 24))

; symbol created to represent a nondeterministic number
(declare-fun |nondet_symex::nondet0| () (_ FloatingPoint 8 24))

; Global guard, used for checking properties
(declare-fun |execution_statet::\guard_exec| () Bool)

; assign the nondeterministic symbol to x
(assert (= |nondet_symex::nondet0| |main::x|))

; assign x to y
(assert (= |main::x| |main::y|))

; assert x == y
(assert (let ((a!1 (not (=> true
                        (=> |execution_statet::\guard_exec|
                          (fp.eq |main::x| |main::y|))))))
  (or a!1))))
```

Assignment of
nondeterministic
value to x



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← Assignment x to y

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```

Check if the comparison
satisfies the guard



```
; assert x == y
(assert (let ((a!1 (not (=> true
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                          (fp.eq |main::x| |main::y|))))))
        (or a!1)))
```

Floating-point Encoding: Illustrative Example

- Z3 produces:

```
sat
(model
  (define-fun |main::x| () (_ FloatingPoint 8 24)
    (_ NaN 8 24))
  (define-fun |main::y| () (_ FloatingPoint 8 24)
    (_ NaN 8 24))
  (define-fun |nondet_symex::nondet0| () (_ FloatingPoint 8 24)
    (_ NaN 8 24))
  (define-fun |execution_statet::\\guard_exec| () Bool
    true)
)
```

Floating-point Encoding: Illustrative Example

- MathSAT produces:

```
sat
( (|main::x| (_ NaN 8 24))
  (|main::y| (_ NaN 8 24))
  (|nondet_symex::nondet0| (_ NaN 8 24))
  (|execution_statet::\guard_exec| true) )
```

Floating-point Encoding: Illustrative Example

Counterexample:

State 1 file main3.c line 3 function main thread 0
main

main3::main::1::x=-NaN (11111111100000000000000000000001)

State 2 file main3.c line 4 function main thread 0
main

main3::main::2::y=-NaN (11111111100000000000000000000001)

State 3 file main3.c line 5 function main thread 0
main

Violated property:
file main3.c line 5 function main
assertion
(_Bool)(x == y)

VERIFICATION FAILED

Encoding of Pointers

- arrays and records / tuples typically handled directly by SMT-solver
- pointers modelled as tuples
 - $p.o \triangleq$ representation of underlying object
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int main() {  
    int a[2], i, x, *p;  
    p=a;  
    if (x==0)  
        a[i]=0;  
    else  
        a[i+1]=1;  
    assert(* (p+2)==1);  
}
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C :=

$$\left(\begin{array}{l} p_1 := \text{store}(p_0, 0, \&a[0]) \\ \wedge p_2 := \text{store}(p_1, 1, 0) \\ \wedge g_2 := (x_2 == 0) \\ \wedge a_1 := \text{store}(a_0, i_0, 0) \\ \wedge a_2 := a_0 \\ \wedge a_3 := \text{store}(a_2, 1 + i_0, 1) \\ \wedge a_4 := \text{ite}(g_1, a_1, a_3) \\ \wedge p_3 := \text{store}(p_2, 1, \text{select}(p_2, 1) + 2) \end{array} \right)$$

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Store object at position 0

```
int main() {  
  int a[2], i, x, *p;  
  p=a;  
  if (x==0)  
    a[i]=0;  
  else  
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  assert(* (p+2)==1);  
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    p=a;
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C :=

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Store index at position 1

Encoding of Pointers

- arrays and records / tuples typically handled directly by SMT-solver
- pointers modelled as tuples
 - $p.o \triangleq$ representation of underlying object
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```
int main() {
  int a[2], i, x, *p;
  p=a;
  if (x==0)
    a[i]=0;
  else
    a[i+1]=1;
  assert(* (p+2)==1);
}
```



C

```

{
  p1 := store(p0, 0, &a[0])
  ∧ p2 := store(p1, 1, 0)
  ∧ g2 := (x2 == 0)
  ∧ a1 := store(a0, i0, 0)
  ∧ a3 := store(a2, 1+ i0, 1)
  ∧ a4 := ite(g1, a1, a3)
  ∧ p3 := store(p2, 1, select(p2, 1)+2)
}
```

*Store object at
position 0*

*Store index at
position 1*

Update index

Encoding of Pointers

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 - p.o $\triangleq$ representation of underlying object
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```
int main() {
    int a[2], i, x, *p;
    p=a;
    if (x==0)
        a[i]=0;
    else
        a[i+1]=1;
    assert(* (p+2)==1);
}
```



P :=

$$\left(\begin{array}{l} i_0 \geq 0 \wedge i_0 < 2 \\ \wedge 1 + i_0 \geq 0 \wedge 1 + i_0 < 2 \\ \wedge \text{select}(p_3, 0) == \&a[0] \\ \wedge \text{select}(\text{select}(p_3, 0), \\ \quad \text{select}(p_3, 1)) == 1 \end{array} \right)$$

*negation satisfiable
(a[2] unconstrained)
 $\Rightarrow$ assert fails*

Encoding of Memory Allocation

- model memory just as an array of bytes (array theories)
 - read and write operations to the memory array on the logic level

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- each dynamic object d_o consists of
 - $m \triangleq$ memory array
 - $s \triangleq$ size in bytes of m
 - $\rho \triangleq$ unique identifier
 - $v \triangleq$ indicate whether the object is still alive
 - $l \triangleq$ the location in the execution where m is allocated

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 - $\rho \triangleq$ unique identifier
 - $v \triangleq$ indicate whether the object is still alive
 - $l \triangleq$ the location in the execution where m is allocated
- to detect invalid reads/writes, we check whether
 - d_o is a dynamic object
 - i is within the bounds of the memory array

$$l_{is_dynamic_object} \Leftrightarrow \left(\bigvee_{j=1}^k d_o.\rho = j \right) \wedge (0 \leq i < n)$$

Encoding of Memory Allocation

- to check for invalid objects, we
 - set v to *true* when the function `malloc` is called (d_o is alive)
 - set v to *false* when the function `free` is called (d_o is not longer alive)

$$I_{valid_object} \Leftrightarrow (I_{is_dynamic_object} \Rightarrow d_o.v)$$

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$$I_{valid_object} \Leftrightarrow (I_{is_dynamic_object} \Rightarrow d_o.v)$$

- to detect forgotten memory, at the end of the (unrolled) program we check
 - whether the d_o has been deallocated by the function `free`

$$I_{deallocated_object} \Leftrightarrow (I_{is_dynamic_object} \Rightarrow \neg d_o.v)$$

Example of Memory Allocation

```
#include <stdio.h>
```

```
void main() {
```

```
    char *p = NULL;
```

```
    char *q = malloc(5); // p = 2
```

```
    p=q;
```

```
    free(p)
```

```
    p = malloc(5);           // p = 3
```

```
    free(p)
```

```
}
```

memory leak: pointer
reassignment makes $d_{o1}.u$
to become an orphan

Example of Memory Allocation

```
#include <stdlib.h>
```

```
void main() {
```

```
    char *p = malloc(5); //  $\rho = 1$ 
```

```
    char *q = malloc(5); //  $\rho = 2$ 
```

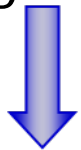
```
    p=q;
```

```
    free(p)
```

```
    p = malloc(5);           //  $\rho = 3$ 
```

```
    free(p)
```

```
}
```



$P := (\neg d_{o1}.v \wedge \neg d_{o2}.v \wedge \neg d_{o3}.v)$

$C := \left(\begin{array}{l} d_{o1}.\rho=1 \wedge d_{o1}.s=5 \wedge d_{o1}.v=true \wedge p=d_{o1} \\ \wedge d_{o2}.\rho=2 \wedge d_{o2}.s=5 \wedge d_{o2}.v=true \wedge q=d_{o2} \\ \wedge p=d_{o2} \wedge d_{o2}.v=false \\ \wedge d_{o3}.\rho=3 \wedge d_{o3}.s=5 \wedge d_{o3}.v=true \wedge p=d_{o3} \\ \wedge d_{o3}.v=false \end{array} \right)$

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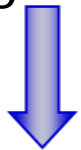
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 - E.g., an integer pointer's value must be aligned to at least 4 bytes, for 32-bit integers

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Align-guaranteed memory mode

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 - E.g., an integer pointer's value must be aligned to at least 4 bytes, for 32-bit integers
- Encode **property assertions** when dereferences occur during symbolic execution
 - To guard against executions where an unaligned pointer is dereferenced
 - This is not as strong as the C standard requirement, that a pointer variable may never hold an unaligned value
 - But it provides a guarantee that any pointer dereference will either be correctly aligned or result in a verification failure

ESBMC's memory model

- statically tracks possible pointer variable targets (objects)
 - dereferencing a pointer leads to the construction of **guarded references** to each **potential target**

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struct foo {  
    uint16_t bar[2];  
    uint8_t baz;  
};
```

```
struct foo qux;  
char *quux = &qux;  
quux++;  
*quux; ← pointer and object types  
do not match
```

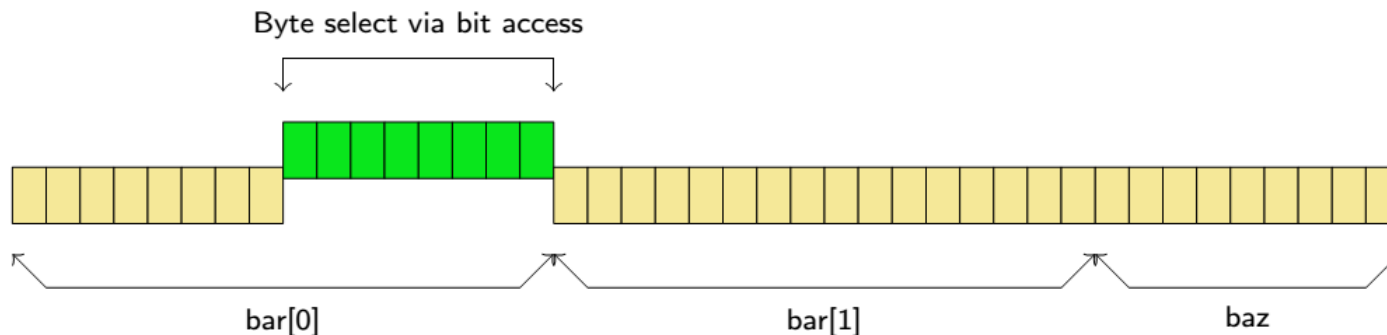

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- SAT: immediate access** to bit-level representation



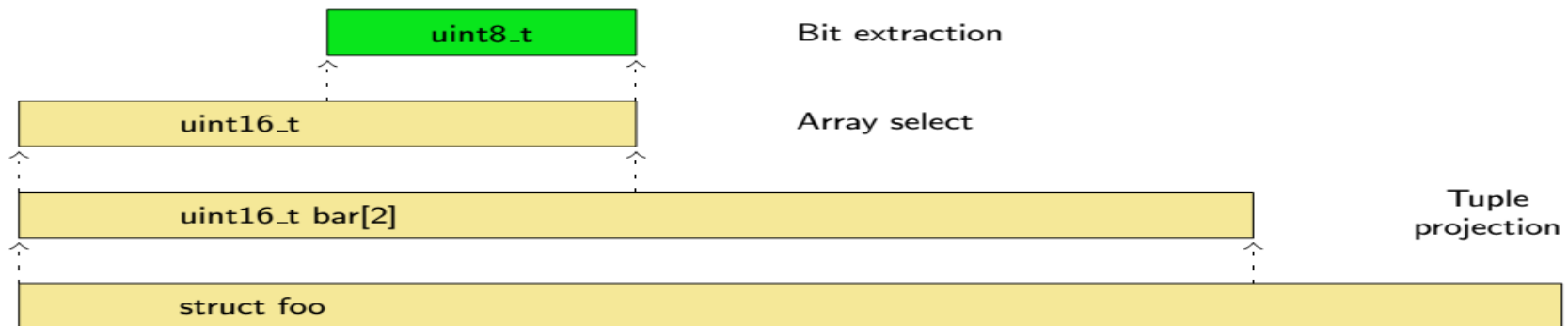
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- SMT:** sorts must be **repeatedly unwrapped**



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- access to underlying data bytes is complicated
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 - requires manipulation of arrays / tuples
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```
uint16_t *fuzz;
```

```
if (nondet_bool()) {
```

```
    fuzz = &qux.bar[0];
```

```
} else {
```

```
    fuzz = &qux.baz;
```

```
}
```

– chooses accessed field nondeterministically

– requires a *byte_extract* expression

– handles the *tuple* that encoded the *struct*

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– requires a *byte_extract* expression
– handles the *tuple* that encoded the *struct*

- supporting **all legal behaviors** at SMT layer **difficult**
 - extract (unaligned) 16bit integer from *fuzz
- experiments showed significantly increased **memory consumption**

“Aligned” Memory Model

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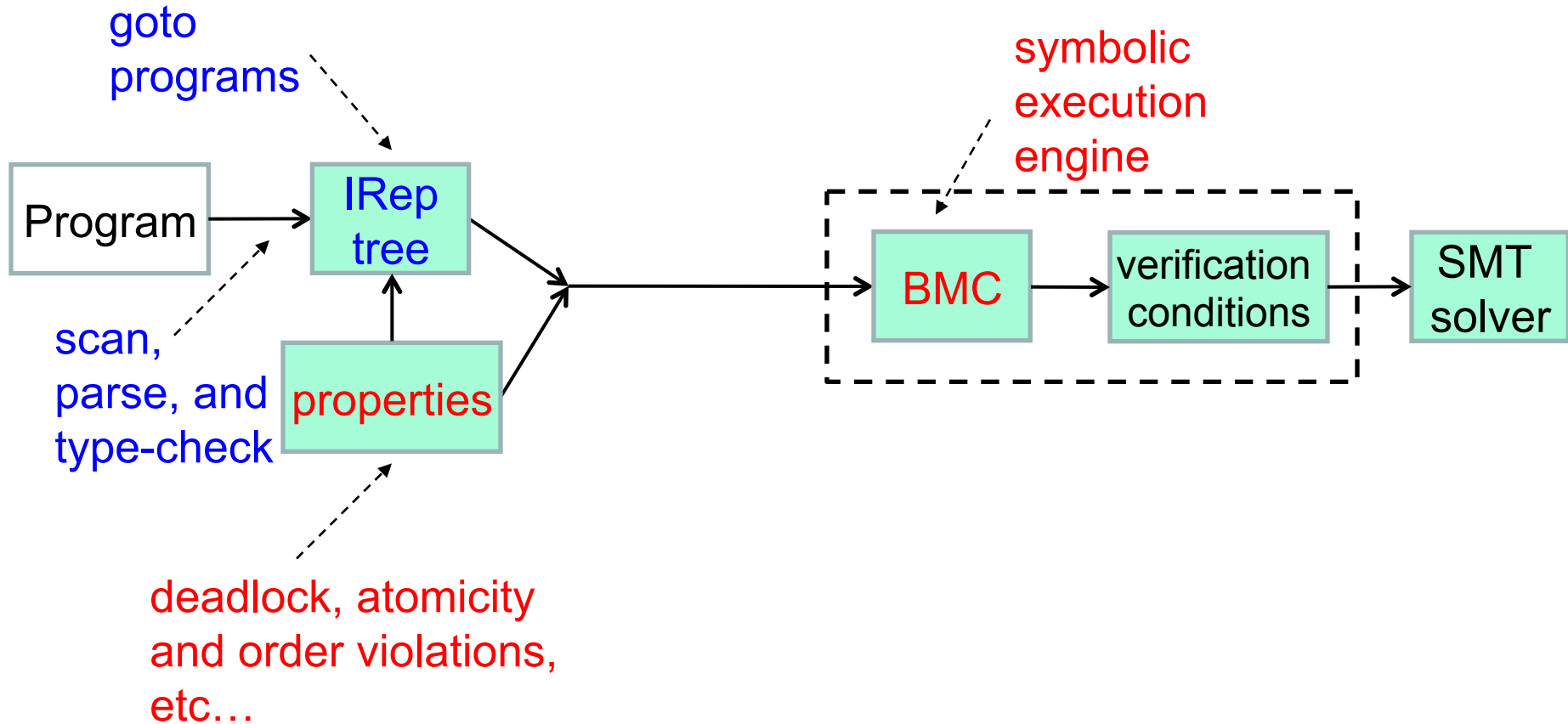
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- **enforce C alignment rules**
 - static analysis of pointer alignment eliminates need to encode unaligned data accesses
 - reduces number of behaviors that must be modeled
 - add alignment assertions (if static analysis not conclusive)
 - extracting 16-bit integer from *fuzz:
 - offset = 0: project bar[0] out of foo
 - offset = 1: **“unaligned memory access” failure**
 - offset = 2: project bar[1] out of foo
 - offset = 3: **“unaligned memory access” failure**
 - offset = 4: “access to object out of bounds” failure

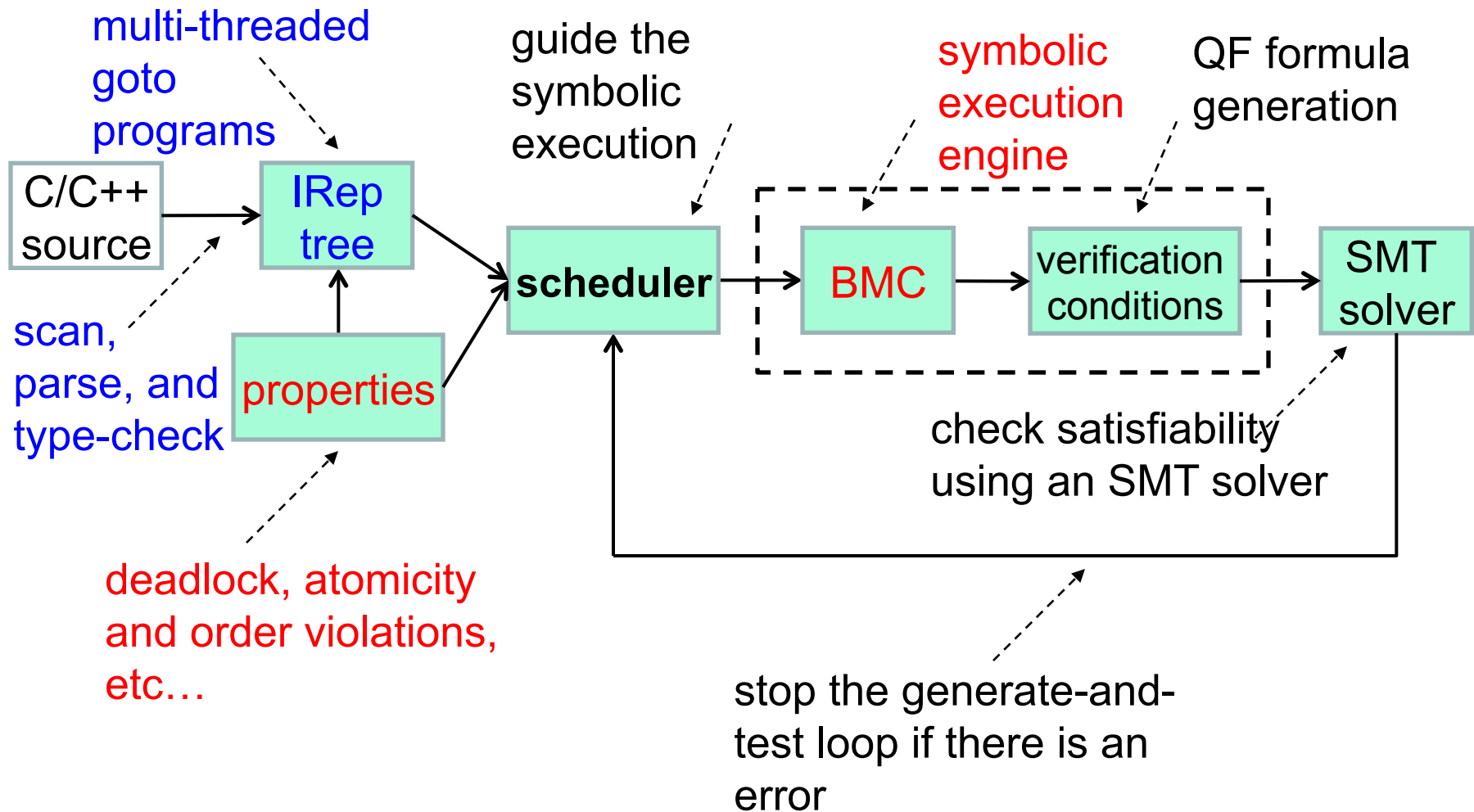
BMC Architecture

- A typical BMC architecture for verifying programs



BMC of Multi-threaded Software

Idea: iteratively generate all possible interleavings and call the BMC procedure on each interleaving



Running Example

- the program has sequences of operations that need to be protected together to avoid atomicity violation
 - requirement: the region of code (val1 and val2) should execute atomically

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

*program counter: 0
mutexes: m1=0; m2=0;
global variables: val1=0; val2=0;
local variables: t1= -1; t2= -1;*

A state $s \in S$ consists of the value of the program counter pc and the values of all program variables

```
7: unlock(m1);  
8: t1 = val1;  
9: lock(m1);  
10: val1 = t1 + 1;  
11: unlock(m1);  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

Lazy exploration: interleaving I_s

statements:

val1-access:

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
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6: unlock(m2);
```

program counter: 0

mutexes: m1=0; m2=0;

global variables: val1=0; val2=0;

local variables: t1= -1; t2= -1;

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

Lazy exploration: interleaving I_s

statements: 1

val1-access:

val2-access:

Thread twoStage

- 1: *lock(m1);*
- 2: *val1 = 1;*
- 3: *unlock(m1);*
- 4: *lock(m2);*
- 5: *val2 = val1 + 1;*
- 6: *unlock(m2);*

program counter: 1

mutexes: m1=1; m2=0;

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Thread reader

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```


Lazy exploration: interleaving I_s

statements: 1-2

val1-access: $W_{\text{twoStage},2}$

val2-access:

write access to the shared variable **val1** in statement **2** of the thread **twoStage**

Thread twoStage

1: lock(m1);

2: **val1 = 1;**

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

program counter: 2

mutexes: m1=1; m2=0;

global variables: **val1=1**; val2=0;

local variables: t1= -1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3

val1-access: $W_{\text{twoStage},2}$

val2-access:

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2 == (t1 + 1));

program counter: 3

mutexes: **m1=0**; m2=0;

global variables: val1=1; val2=0;

local variables: t1= -1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7

val1-access: $W_{\text{twoStage},2}$

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

program counter: 7

mutexes: **m1=1**; m2=0;

global variables: val1=1; val2=0;

local variables: t1= -1; t2= -1;

Lazy exploration: interleaving L

statements: 1-2-3-7-8

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8}$

val2-access:

read access to the shared variable *val1* in statement 8 of the thread *reader*

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

program counter: 8

mutexes: m1=1; m2=0;

global variables: val1=1; val2=0;

local variables: t1= -1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11}$

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

program counter: 11

mutexes: m1=1; m2=0;

global variables: val1=1; val2=0;

local variables: **t1= 1**; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11}$

val2-access:

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

● 12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

program counter: 12

mutexes: **m1=0**; m2=0;

global variables: val1=1; val2=0;

local variables: t1= 1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11}$

val2-access:

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

CS2

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

program counter: 4

mutexes: m1=0; m2=0;

global variables: val1=1; val2=0;

local variables: t1= 1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11}$

val2-access:

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

CS2

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2 == (t1 + 1));

program counter: 4

mutexes: m1=0; **m2=1**;

global variables: val1=1; val2=0;

local variables: t1= 1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11} - R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

CS2

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

● **program counter: 5**

mutexes: m1=0; m2=1;

global variables: val1=1; **val2=2;**

local variables: t1= 1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11} - R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

● 6: unlock(m2);

CS1

CS2

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

program counter: 6

mutexes: m1=0; **m2=0**;

global variables: val1=1; val2=2;

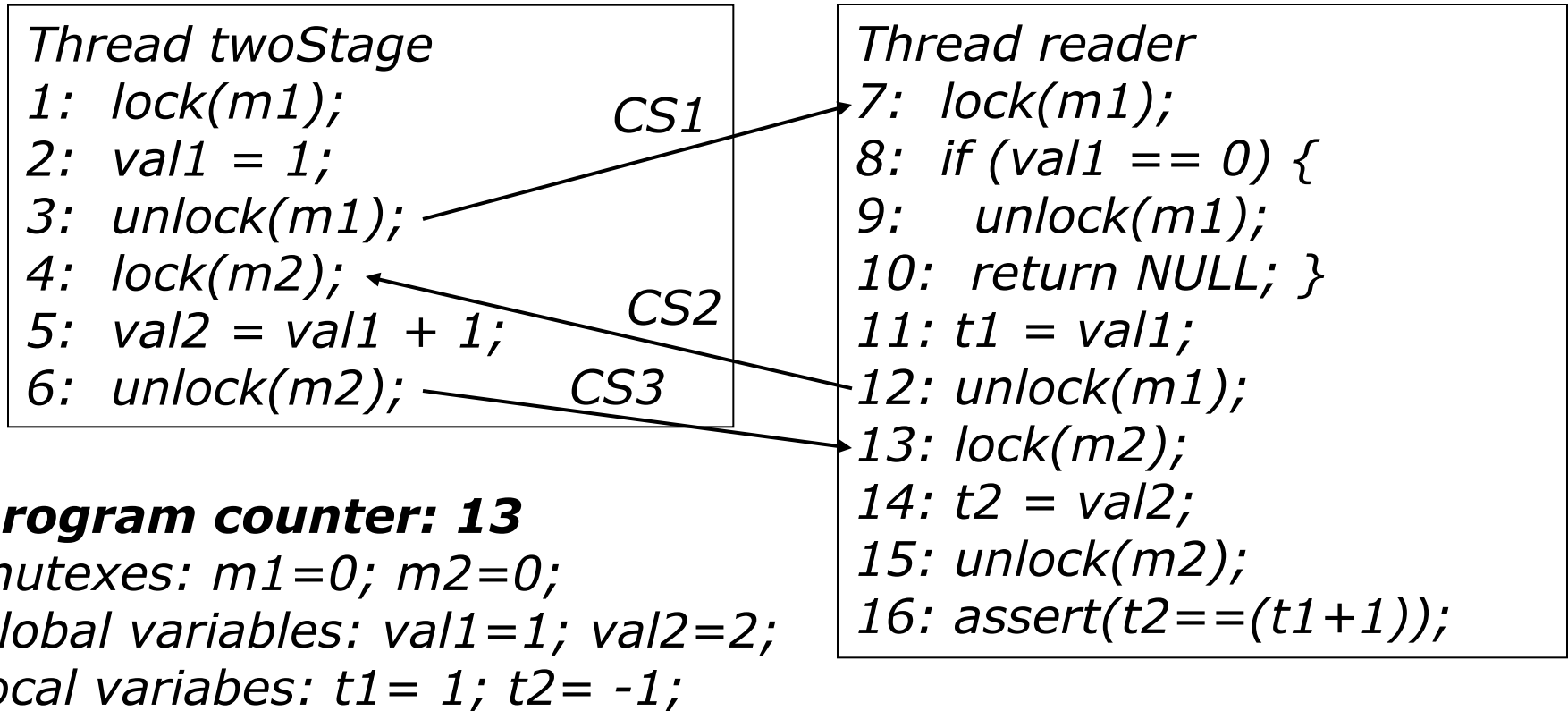
local variables: t1= 1; t2= -1;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11} - R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$

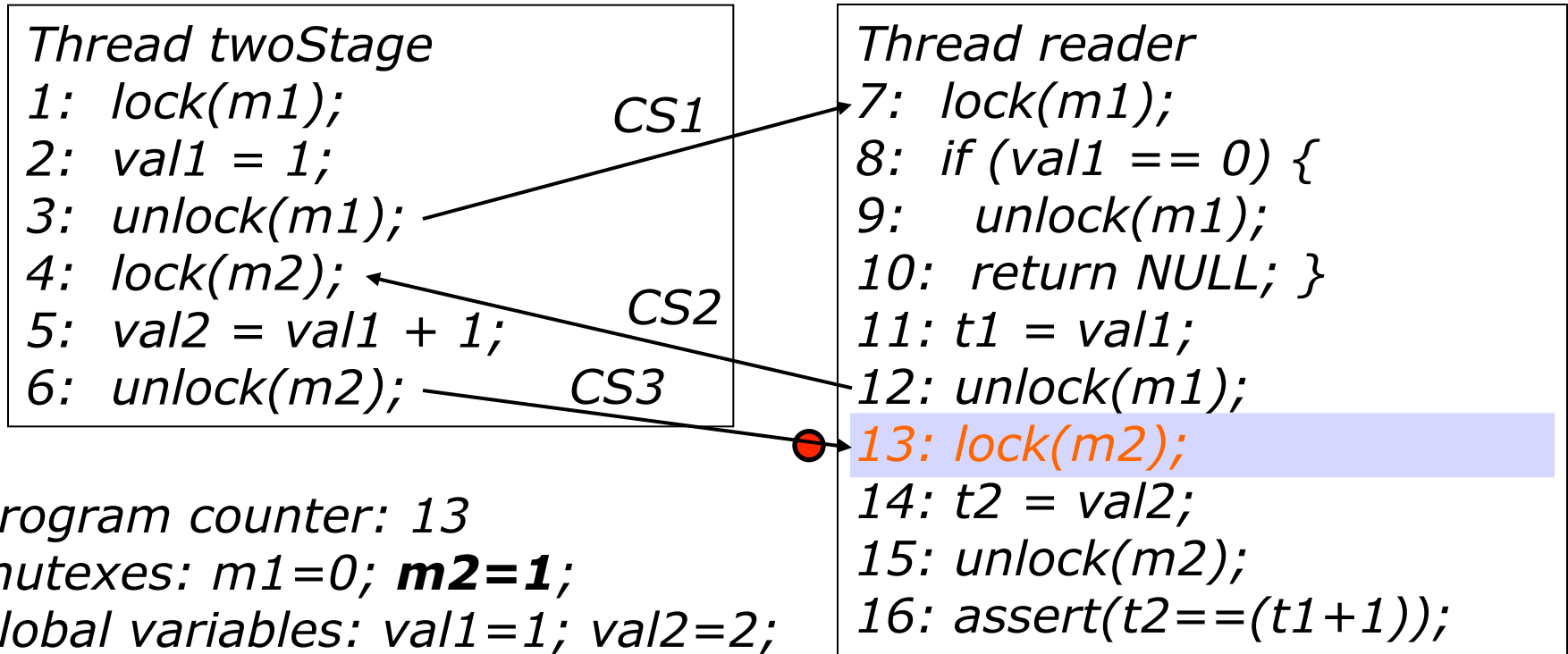


Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6-13

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11} - R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$



program counter: 13

mutexes: $m1=0$; **$m2=1$** ;

global variables: $val1=1$; $val2=2$;

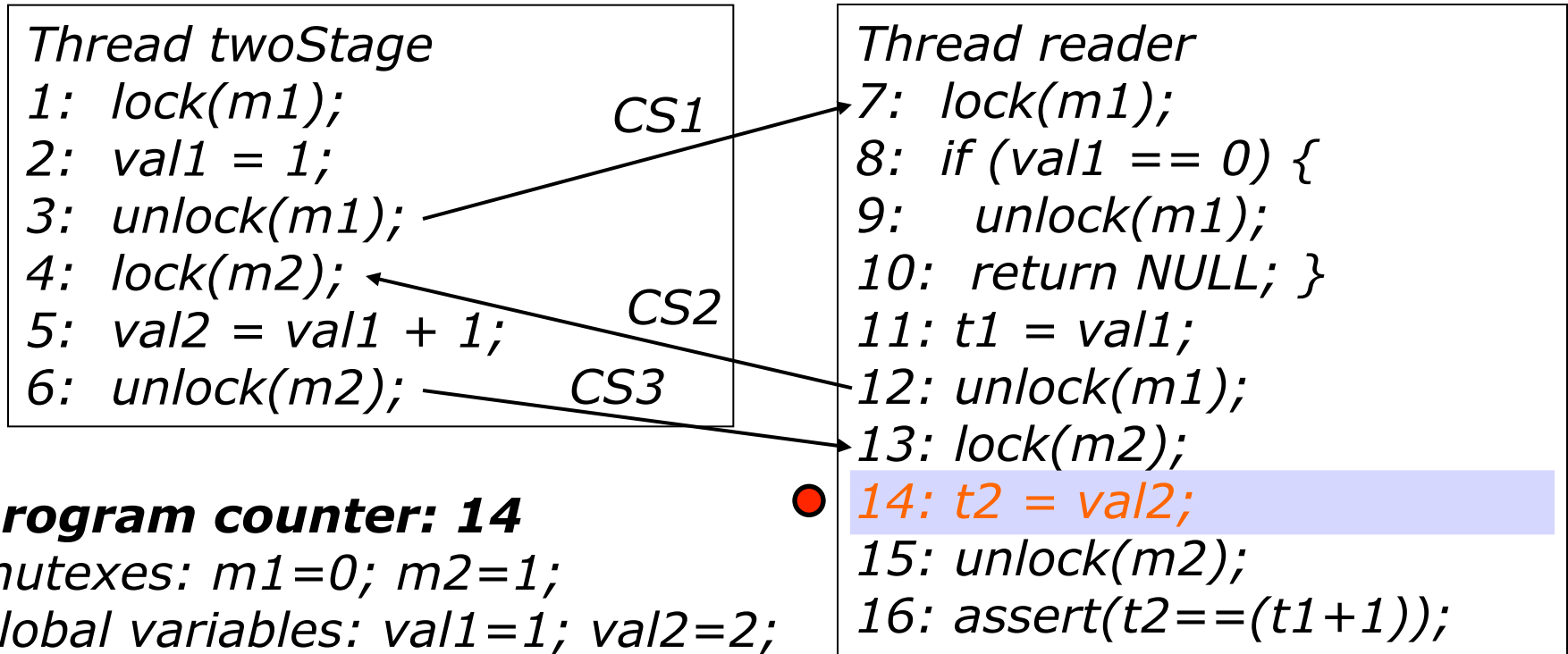
local variables: $t1=1$; $t2=-1$;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6-13-14

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$ - $R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$ - $R_{\text{reader},14}$



Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6-13-14-15

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$ - $R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$ - $R_{\text{reader},14}$

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

CS2

CS3

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

program counter: 15

mutexes: m1=0; **m2=0**;

global variables: val1=1; val2=2;

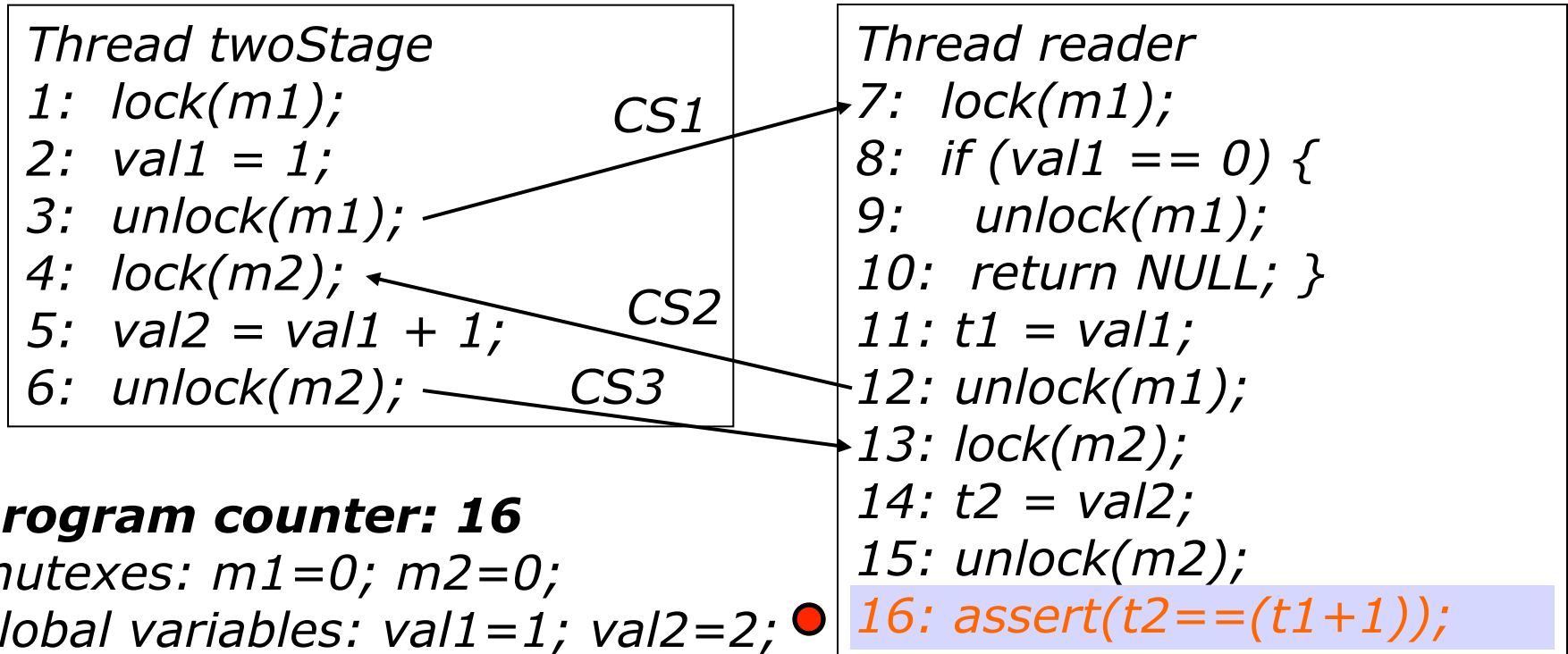
local variables: t1= 1; t2= 2;

Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6-13-14-15-16

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$ - $R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$ - $R_{\text{reader},14}$



Lazy exploration: interleaving I_s

statements: 1-2-3-7-8-11-12-4-5-6-13-14-15-16

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$ - $R_{\text{twoStage},5}$

val2-access: $W_{\text{twoStage},5}$ - $R_{\text{reader},14}$

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

CS2

CS3

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

QF formula is unsatisfiable,
i.e., assertion holds

Lazy exploration: interleaving I_f

statements:

val1-access:

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

program counter: 0

mutexes: m1=0; m2=0;

global variables: val1=0; val2=0;

local variables: t1= -1; t2= -1;

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

Lazy exploration: interleaving I_f

statements: 1-2-3

val1-access: $W_{\text{twoStage},2}$

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

program counter: 3

mutexes: m1=0; m2=0;

global variables: **val1=1**; val2=0;

local variables: t1= -1; t2= -1;

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

Lazy exploration: interleaving I_f

statements: 1-2-3

val1-access: $W_{\text{twoStage},2}$

val2-access:

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

program counter: 7

mutexes: m1=0; m2=0;

global variables: val1=1; val2=0;

local variables: t1= -1; t2= -1;

Lazy exploration: interleaving I_f

statements: 1-2-3-7-8-11-12-13-14-15-16

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11}$

val2-access: $R_{\text{reader},14}$

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2==(t1+1));
```

program counter: 16

mutexes: $m1=0$; $m2=0$;

global variables: $val1=1$; $val2=0$;

local variables: **$t1=1$** ; **$t2=0$** ;

Lazy exploration: interleaving I_f

statements: 1-2-3-7-8-11-12-13-14-15-16

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$

val2-access: $R_{\text{reader},14}$

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2 == (t1 + 1));
```

CS2

program counter: 4

mutexes: $m1=0$; $m2=0$;

global variables: $val1=1$; $val2=0$;

local variables: $t1=1$; $t2=0$;

Lazy exploration: interleaving I_f

statements: 1-2-3-7-8-11-12-13-14-15-16-4-5-6

val1-access: $W_{\text{twoStage},2}$ - $R_{\text{reader},8}$ - $R_{\text{reader},11}$ - $R_{\text{twoStage},5}$

val2-access: $R_{\text{reader},14}$ - $W_{\text{twoStage},5}$

Thread twoStage

1: lock(m1);

2: val1 = 1;

3: unlock(m1);

4: lock(m2);

5: val2 = val1 + 1;

6: unlock(m2);

CS1

Thread reader

7: lock(m1);

8: if (val1 == 0) {

9: unlock(m1);

10: return NULL; }

11: t1 = val1;

12: unlock(m1);

13: lock(m2);

14: t2 = val2;

15: unlock(m2);

16: assert(t2==(t1+1));

CS2

program counter: 6

mutexes: m1=0; m2=0;

global variables: val1=1; **val2=2**;

local variables: t1= 1; t2= 0;

Lazy exploration: interleaving I_f

statements: 1-2-3-7-8-11-12-13-14-15-16-4-5-6

val1-access: $W_{\text{twoStage},2} - R_{\text{reader},8} - R_{\text{reader},11} - R_{\text{twoStage},5}$

val2-access: $R_{\text{reader},14} - W_{\text{twoStage},5}$

Thread twoStage

```
1: lock(m1);  
2: val1 = 1;  
3: unlock(m1);  
4: lock(m2);  
5: val2 = val1 + 1;  
6: unlock(m2);
```

CS1

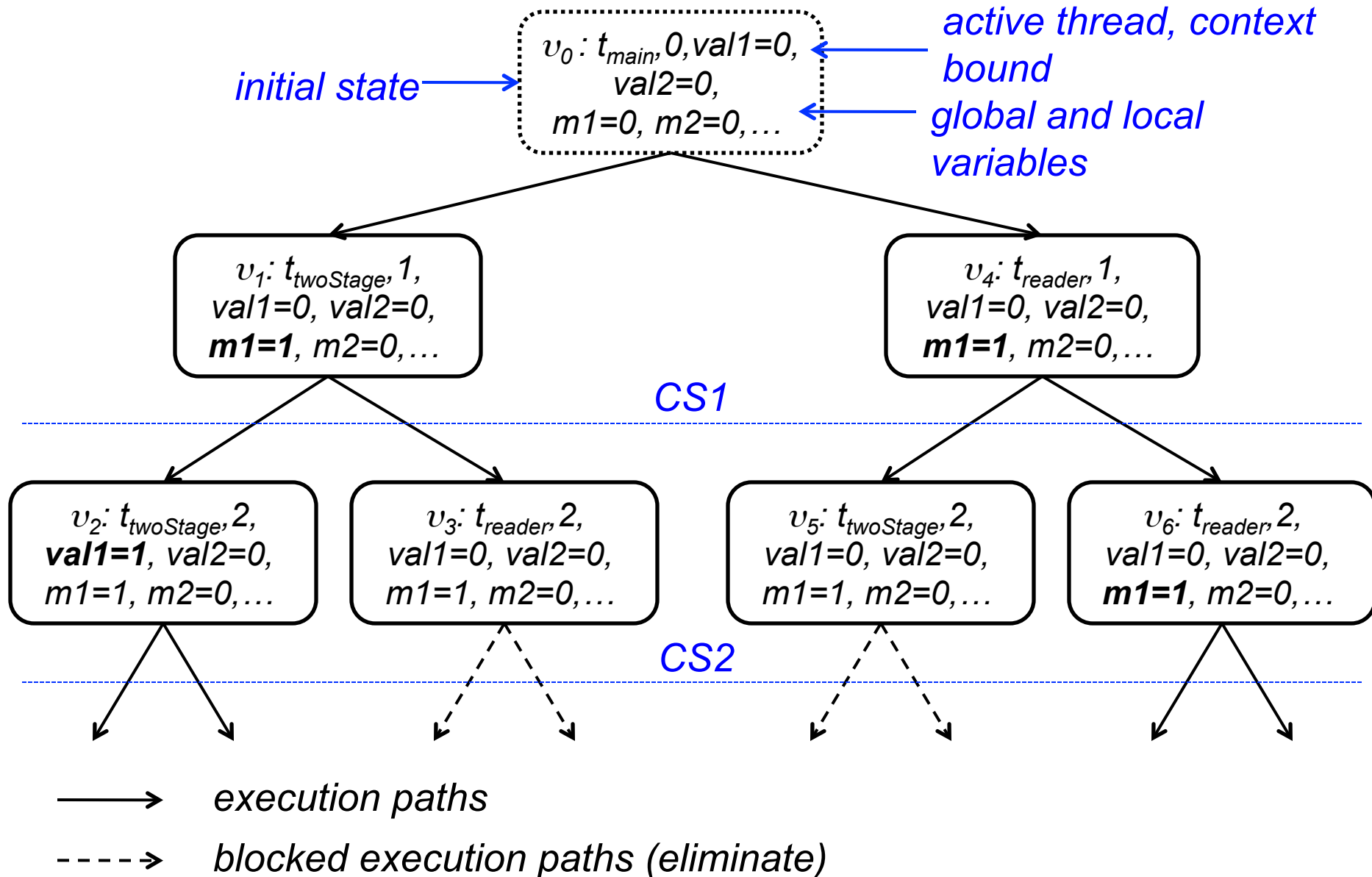
Thread reader

```
7: lock(m1);  
8: if (val1 == 0) {  
9:   unlock(m1);  
10:  return NULL; }  
11: t1 = val1;  
12: unlock(m1);  
13: lock(m2);  
14: t2 = val2;  
15: unlock(m2);  
16: assert(t2 == (t1 + 1));
```

CS2

*QF formula is satisfiable,
i.e., assertion does not hold*

Lazy Approach: State Transitions



Exploring the Reachability Tree

- Use a reachability tree (RT) to describe reachable states of a multi-threaded program

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- Each node in the RT is a tuple $v = \left(A_i, C_i, s_i, \left\langle l_i^j, G_i^j \right\rangle_{j=1}^n \right)_i$ for a given time step i , where:

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 - A_i represents the currently active thread

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 - A_i represents the currently active thread
 - C_i represents the context switch number

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 - C_i represents the context switch number
 - s_i represents the current state

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 - A_i represents the currently active thread
 - C_i represents the context switch number
 - s_i represents the current state
 - l_i^j represents the current location of thread j

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 - A_i represents the currently active thread
 - C_i represents the context switch number
 - s_i represents the current state
 - l_i^j represents the current location of thread j
 - G_i^j represents the control flow guards accumulated in thread j along the path from l_0^j to l_i^j

Expansion Rules of the RT

R1 (assign): If l is an assignment, we execute l , which generates s_{i+1} . We add as child to v a new node v'

$$v' = \left(\underline{A_i}, \underline{C_i}, \underline{s_{i+1}}, \left\langle \underline{l_{i+1}^j}, G_i^j \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^{A_i} = l_i^{A_i} + 1$$

- we have fully expanded v if
 - l within an atomic block; or
 - l contains no global variable; or
 - the upper bound of context switches ($C_i = C$) is reached
- if v is not fully expanded, for each thread $j \neq A_i$ where G_i^j is enabled in s_{i+1} , we thus create a new child node

$$v'_j = \left(\underline{j}, \underline{C_i + 1}, \underline{s_{i+1}}, \left\langle l_i^j, G_i^j \right\rangle \right)_{i+1}$$

Expansion Rules of the RT

R2 (skip): If l is a *skip*-statement with target l , we increment the location of the current thread and continue with it. We explore no context switches:

$$v' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, G_i^j \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^j = \begin{cases} l_i^j + 1 & : j = A_i \\ l_i^j & : \text{otherwise} \end{cases}$$

Expansion Rules of the RT

R2 (skip): If l is a *skip*-statement with target l , we increment the location of the current thread and continue with it. We explore no context switches:

$$v' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, G_i^j \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^j = \begin{cases} l_i^j + 1 & : j = A_i \\ l_i^j & : \text{otherwise} \end{cases}$$

R3 (unconditional goto): If l is an unconditional *goto*-statement with target l , we set the location of the current thread and continue with it. We explore no context switches:

$$v' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, G_i^j \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^j = \begin{cases} l & : j = A_i \\ l_i^j & : \text{otherwise} \end{cases}$$

Expansion Rules of the RT

R4 (conditional goto): If l is a conditional *goto*-statement with test c and target l , we create two child nodes v' and v'' .

- for v' , we assume that c is *true* and proceed with the target instruction of the jump:

$$v' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, \underline{c \wedge G_i^j} \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^j = \begin{cases} l & : j = A_i \\ l_i^j & : \text{otherwise} \end{cases}$$

- for v'' , we add $\neg c$ to the guards and continue with the next instruction in the current thread

$$v'' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, \underline{\neg c \wedge G_i^j} \right\rangle \right)_{i+1} \xrightarrow{\quad} l_{i+1}^j = \begin{cases} l_i^j + 1 & : j = A_i \\ l_i^j & : \text{otherwise} \end{cases}$$

- prune one of the nodes if the condition is determined statically

Expansion Rules of the RT

R5 (assume): If l is an *assume*-statement with argument c , we proceed similar to R1.

- we continue with the unchanged state s_i but add c to all guards, as described in R4
- If $c \wedge G_i^j$ evaluates to *false*, we prune the execution path

Expansion Rules of the RT

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- If $c \wedge G_i^j$ evaluates to *false*, we prune the execution path

R6 (assert): If l is an *assert*-statement with argument c , we proceed similar to R1.

- we continue with the unchanged state s_i but add c to all guards, as described in R4
- we generate a verification condition to check the validity of c

Expansion Rules of the RT

R5 (start_thread): If l is a *start_thread* instruction, we add the indicated thread to the set of active threads:

$$v' = \left(A_i, C_i, s_i, \underbrace{\left\langle l_{i+1}^j, G_{i+1}^j \right\rangle}_{j=1}^{n+1} \right)_{i+1}$$

- where l_{i+1}^{n+1} is the initial location of the thread and $G_{i+1}^{n+1} = G_i^{A_i}$
- the thread starts with the guards of the currently active thread

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- the thread starts with the guards of the currently active thread

R6 (join_thread): If l is a *join_thread* instruction with argument ld , we add a child node:

$$v' = \left(A_i, C_i, s_i, \left\langle \underline{l_{i+1}^j}, G_i^j \right\rangle \right)_{i+1}$$

- where $l_{i+1}^j = l_i^{A_i} + 1$ only if the joining thread ld has exited

Lazy exploration of interleavings

- Main steps of the algorithm:
 1. Initialize the stack with the initial node v_0 and the initial path $\pi_0 = \langle v_0 \rangle$
 2. If the stack is empty, terminate with “no error”.
 3. Pop the current node v and current path π off the stack and compute the set v' of successors of v using rules R1-R8.
 4. If v' is empty, derive the VC φ_k^π for π and call the SMT solver on it. If φ_k^π is satisfiable, terminate with “error”; otherwise, goto step 2.
 5. If v' is not empty, then for each node $v \in v'$, add v to π , and push node and extended path on the stack. goto step 3.

computation path

$$\pi = \{v_1, \dots, v_n\}$$

$$\varphi_k^\pi = \overbrace{I(s_0) \wedge R(s_0, s_1) \wedge \dots \wedge R(s_{k-1}, s_k)}^{\text{constraints}} \wedge \overbrace{\neg \phi_k}^{\text{property}}$$

bound

Observations about the lazy approach

- naïve but useful:
 - bugs usually manifest with few context switches
[Qadeer&Rehof'05]

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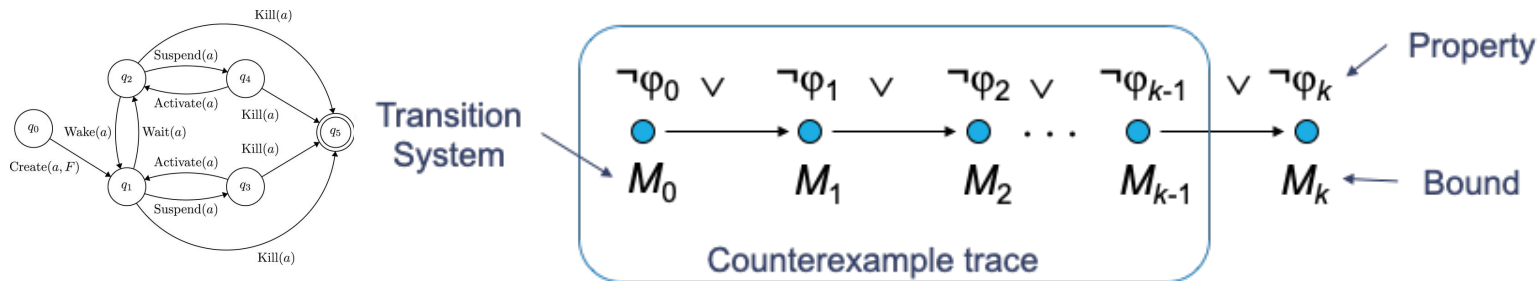
- naïve but useful:
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 - bound the number of preemptions (C) allowed per threads
 - ▷ *number of executions: $O(n^C)$*
 - as each formula corresponds to one possible path only, its size is relatively small
- can suffer performance degradation:
 - in particular for correct programs where we need to invoke the SMT solver once for each possible execution path

Intended learning outcomes

- Understand **soundness** and **completeness** concerning **detection techniques**
- Emphasize the difference between **static analysis** and **testing / simulation**
- Explain **bounded model checking** of software
- Explain **unbounded model checking** of **software**

Revisiting BMC

- Basic Idea: given a transition system M , check negation of a given property φ up to given depth k :



- Translated into a VC ψ such that: ψ satisfiable iff φ has counterexample of max. depth k

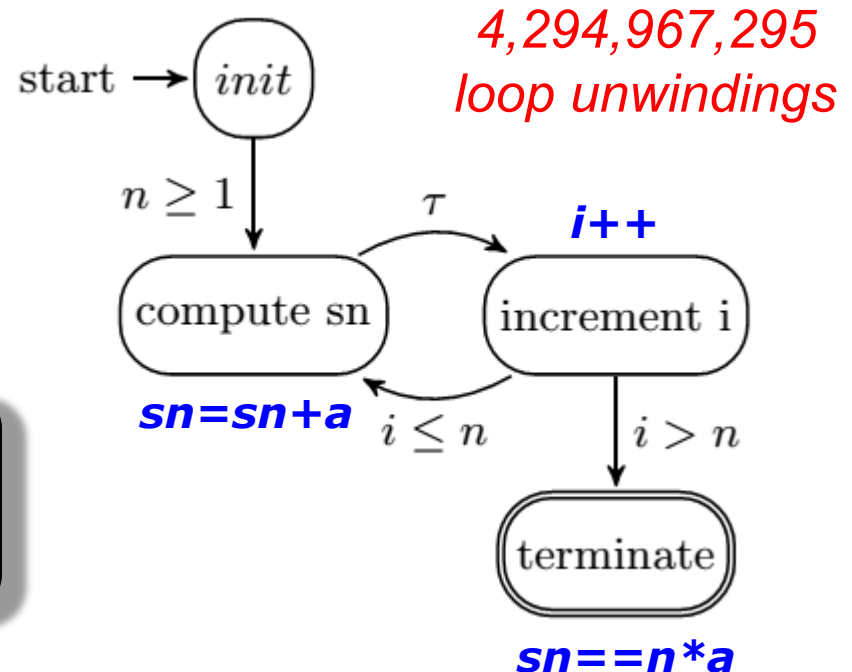
BMC is aimed at finding bugs; it cannot prove correctness, unless the bound k safely reaches all program states

Difficulties in proving the correctness of programs with loops in BMC

- BMC techniques can falsify properties up to a given depth k
 - they can prove correctness only if an upper bound of k is known (**unwinding assertion**)
 - » BMC tools typically fail to verify programs that contain bounded and unbounded loops

$$S_n = \sum_{i=1}^n a = na, n \geq 1$$

the loop will be unfolded 2^{n-1} times
(in the worst case, 2^{32-1} times on 32
bits integer)



Induction-Based Verification

k -induction checks...

- **base case** ($base_k$): find a counter-example with up to k loop unwindings (plain BMC)
- **forward condition** (fwd_k): check that P holds in all states reachable within k unwindings
- **inductive step** ($step_k$): check that whenever P holds for k unwindings, it also holds after next unwinding
 - havoc state
 - run k iterations
 - assume invariant
 - run final iteration

⇒ iterative deepening if inconclusive

The k -induction algorithm

k =initial bound

```
while true do  
  if  $base_k$  then  
    return trace  $s[0..k]$   
  else if  $fwd_k$   
    return true  
  else if  $step_k$  then  
    return true  
  end if  
   $k=k+1$   
end
```

The k -induction algorithm

k =initial bound

while *true* **do**

if $base_k$ **then**

return *trace* $s[0..k]$

else if fwd_k

return *true*

else if $step_k$ **then**

return *true*

end if

$k=k+1$

end

inserts unwinding
assumption after
each loop

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inserts unwinding
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inserts unwinding
assertion after each
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havoc variables that
occur in the loop's
termination condition

unable to falsify or
prove the property

Running example

Prove that $S_n = \sum_{i=1}^n a = na$ for $n \geq 1$

```
unsigned int nondet_uint();  
int main() {  
    unsigned int i, n=nondet_uint(), sn=0;  
    assume (n>=1);  
    for(i=1; i<=n; i++)  
        sn = sn + a;  
    assert(sn==n*a);  
}
```


Running example: *base case*

Insert an **unwinding assumption** consisting of the termination condition after the loop

- find a counter-example with k loop unwindings

```
unsigned int nondet_uint();  
int main() {  
    unsigned int i, n=nondet_uint(), sn=0;  
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    for(i=1; i<=n; i++)  
        sn = sn + a;  
    assume(i>n);  
    assert(sn==n*a);  
}
```

Running example: *forward condition*

Insert an **unwinding assertion** consisting of the termination condition after the loop

- check that P holds in all states reachable with k unwindings

```
unsigned int nondet_uint();  
int main() {  
    unsigned int i, n=nondet_uint(), sn=0;  
    assume (n>=1);  
    for(i=1; i<=n; i++)  
        sn = sn + a;  
    assert(i>n);  
    assert(sn==n*a);  
}
```

Running example: *inductive step*

Havoc (only) the variables that occur in the loop's termination and branch conditions

```
unsigned int nondet_uint();  
typedef struct state {  
    unsigned int i, n, sn;  
} statet;  
int main() {  
    unsigned int i, n=nondet_uint(), sn=0, k;  
    assume(n>=1);  
    statet cs, s[n];  
    cs.i=nondet_uint();  
    cs.sn=nondet_uint();  
    cs.n=n;
```

Running example: *inductive step*

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define the type of the
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```

define the type of the
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state vector

explore all possible
values implicitly

Running example: *inductive step*

BMC is called to verify the assertions where the first arbitrary state is emulated by **nondeterminism**

```
for(i=1; i<=n; i++) {  
    s[i-1]=cs;  
    sn = sn + a;  
    cs.i=i;  
    cs.sn=sn;  
    cs.n=n;  
    assume(s[i-1]!=cs);  
}  
assume(i>n);  
assert(sn == n*a);  
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```

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capture the state *cs*
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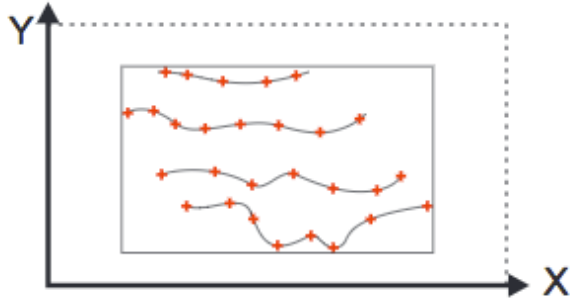
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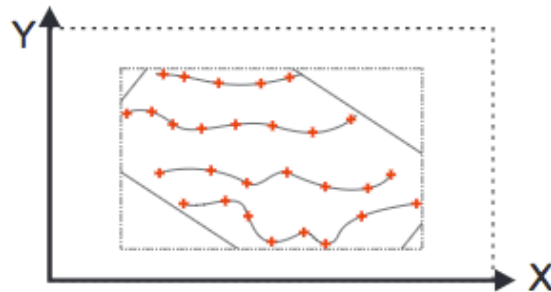
insert unwinding
assumption

Automatic Invariant Generation

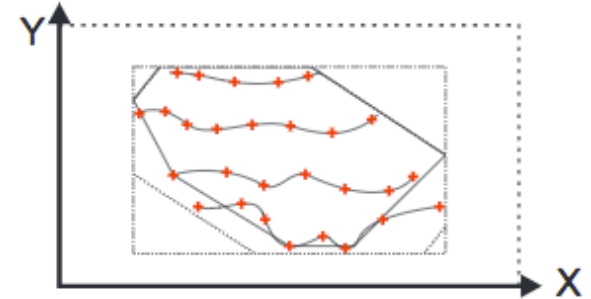
- Infer invariants using **intervals**, **octagons**, and **convex polyhedral** constraints for the inductive step
 - e.g., $a \leq x \leq b$; $x \leq a$, $x - y \leq b$; and $ax + by \leq c$



intervals



octagons



convex polyhedral

- Use existing libraries to discover linear/polynomial relations among integer/real variables to infer **loop invariants**
 - compute **pre-** and **post-conditions**

Running Example: Plain BMC

- Plain BMC unrolls this *while*-loop 100 times...

```
int main() {  
    int x=0, t=0, phase=0;  
    while(t<100) {  
        if(phase==0) x=x+2;  
        if(phase==1) x=x-1;  
        phase=1-phase;  
        t++;  
    }  
    assert(x<=100);  
    return 0;  
}
```

```
$esbmc example.c --clang-frontend  
ESBMC version 4.2.0 64-bit x86_64 macos  
file example.c: Parsing  
Converting  
Type-checking example  
Generating GOTO Program  
GOTO program creation time: 0.232s  
GOTO program processing time: 0.001s  
Starting Bounded Model Checking  
Unwinding loop 1 iteration 1 file example.c line 5 function  
main  
Unwinding loop 1 iteration 2 file example.c line 5 function  
main  
...  
Unwinding loop 1 iteration 100 file example.c line 5 function  
main  
Symex completed in: 0.340s (313 assignments)  
Slicing time: 0.000s  
Generated 1 VCC(s), 0 remaining after simplification  
VERIFICATION SUCCESSFUL  
BMC program time: 0.340s
```

Running Example: *k*-induction + invariants

- Inductive step proves correctness for *k*-step 2...

```
int main() {  
  int x=0, t=0, phase=0;  
  while(t<100) {  
    assume(-2*x+t+3*phase == 0);  
    assume(3-2*x+t >= 0);  
    assume(-x+2*t >= 0);  
    assume(147+x-2*t >= 0);  
    assume(2*x-t >= 0);  
    if(phase==0) x=x+2;  
    if(phase==1) x=x-1;  
    phase=1-phase;  
    t++;  
  }  
  assert(x<=100);  
  return 0;  
}
```

```
$esbmc example.c --clang-frontend --k-induction
```

```
*** K-Induction Loop Iteration 2 ***
```

```
*** Checking inductive step
```

```
Starting Bounded Model Checking
```

```
Unwinding loop 1 iteration 1 file example_pagai.c line 6 function main
```

```
Unwinding loop 1 iteration 2 file example_pagai.c line 6 function main
```

```
Symex completed in: 0.002s (53 assignments)
```

```
Slicing time: 0.000s
```

```
Generated 1 VCC(s), 1 remaining after simplification
```

```
No solver specified; defaulting to Boolector
```

```
Encoding remaining VCC(s) using bit-vector arithmetic
```

```
Encoding to solver time: 0.001s
```

```
Solving with solver Boolector 2.4.0
```

```
Encoding to solver time: 0.001s
```

```
Runtime decision procedure: 0.144s
```

```
VERIFICATION SUCCESSFUL
```

```
BMC program time: 0.148s
```

```
Solution found by the inductive step (k = 2)
```

inductive invariants

reuse *k*-induction counterexamples to speed-up bug finding
reuse results of previous steps (caching SMT queries)

Summary

- Described the difference between **soundness** and **completeness** concerning **detection techniques**
 - **False positive** and **false negative**
- Pointed out the difference between **static analysis** and **testing / simulation**
 - **hybrid combination** of static and dynamic analysis techniques to achieve a good trade-off between **soundness** and **completeness**
- Explained **bounded** and **unbounded model checking of software**
 - they have been applied successfully to verify **single- and multi-threaded software**